Aerial Bordering

The Role of Frontex's Aerial Surveillance in the Central Mediterranean Pushback Machine

This report examines how aerial surveillance has reshaped border control in the Central Mediterranean, enabling widespread violations of migrants' rights. In particular, it shows how, since 2017, Frontex—the European Border and Coast Guard Agency—has progressively built up an extensive fleet of manned and unmanned aircraft operated by private companies, playing a crucial role in facilitating the interception of migrants at sea and their forced return to Libya and Tunisia, where they face grave abuses. Accompanying the database published on LIMINAL's online platform, the report traces the emergence of this phenomenon, unpacks its current operational logics and its contribution to human rights violations, and situates it within broader transformations in bordering practices at the global level.

LIMINAL
INVESTIGATING IM–MOBILITIES AND BORDER VIOLENCE

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Key takeaways

1) What is aerial bordering?
An extensive programme of surveillance from the sky initiated by European authorities at the EU's external borders. This platform focuses specifically on aerial surveillance carried out in the Central Mediterranean by Frontex, the EU border control agency, since 2017. This is when, after years of testing, the Agency started to conduct regular aerial patrols through aircraft operated by private companies capable of streaming real-time video.

2) What does our analysis reveal?
This platform presents the first systematic study of aerial bordering operations in the Central Mediterranean over such an extended period. Due to its scope and use of data collected by the Agency itself, our analysis allows for an assessment of the actual effects of aerial bordering on the ground, beyond institutional narratives and PR strategies. Our analysis shows that:

a. Frontex's aerial bordering programme has grown exponentially over the last decade. This is visible both in the rising amount of money the Agency spends on procuring and testing aerial surveillance services from private contractors (from 11.7€ million in 2018 to 278.6€ million in 2026), and in the increase in the number of hours these planes and drones have been flying in operations over the Central Mediterranean (from 469 hours in 2019 to at least 5253 hours four years later).

b. Between August 2018 and October 2024, Frontex's aircraft detected at least 611 migrant boats carrying a total of **33,107** passengers who were intercepted and forcibly returned to Libya or Tunisia. In 2021, the year with the most complete data available, **detection by Frontex's aircraft accounted for at least 36% of all recorded pushbacks.** This should be understood as a lower-bound estimate: recently released data suggest that Frontex was in fact responsible for approximately half of all pushbacks. Moreover, while Frontex has refused to share data that would allow us to conduct the same analysis on more recent periods, this data further indicates that this proportion applies also to 2026. Overall, this highlights the **Agency's continued and significant role in facilitating forced returns.**

c. This platform focuses on the role of aerial surveillance in what we refer to as the **"pushback machine"** by highlighting **three key operational patterns:** Frontex's communication with actors at sea is selective and consistently prioritises the intervention of Libyan and Tunisian authorities over non-governmental rescue groups; Frontex's aerial operations have resulted in a significant extension of the Libyan Coast Guard's operational area, including areas well into the Maltese Search and Rescue Zone; and that Frontex aircraft regularly witness direct violence by Libyan and Tunisian actors at sea, yet this has not led the Agency to reconsider its role and activities.

d. While Frontex has repeatedly maintained that its aerial surveillance programme is employed in the service of rescue, the Agency's own data, as analysed on this platform, suggests otherwise. In 2021, the year with the most complete data available, **only 5% of operations at sea initiated by a Frontex aircraft sighting led to what, according to current legal standards, can legitimately be defined as rescue** (i.e., when passengers are safely disembarked in Italy or Malta). Beyond search-and-rescue, the Agency characterises its aerial surveillance programme as "multipurpose", including activities such as "fisheries control." Our analysis reveals that this is a mischaracterisation at best, as **border surveillance activities account for 95% of all events recorded by the Agency itself.**

You can read more about how we conducted our data analysis in Methodology

3) Why is aerial bordering a problem?
Our platform reveals how aerial surveillance has become a key component of the EU's strategy to prevent migrants from reaching Europe by boat and to return them with impunity to violence and abuse in Libya and Tunisia. More generally, as unpacked in more detail in our Conclusion , aerial bordering is at the forefront of some of the most concerning developments in border surveillance across the world: the privatisation of border control; the militarisation of borders; and the collusion with the (Israeli) military-industrial complex.

Introduction

On 15 June 2017, Frontex, the European Border and Coast Guard Agency, conducted the first operational flight under its newly established Multipurpose Aerial Surveillance (MAS) service. Following a long testing phase, a surveillance plane equipped with a camera and other sensors live-streamed full motion video and other data from the Central Mediterranean to Frontex headquarters in Warsaw and other coordination centres, where duty officers from multiple European authorities assessed the information.

Operational data collected by Frontex itself and the analysis we provide in this platform show that these flights, which have since become increasingly frequent, are overwhelmingly dedicated to one task: spotting migrant boats to enable their interception and forcible return to Libya and Tunisia in full knowledge of the systemic violence and abuse to which passengers of these boats are subjected in those countries.

The mid-June flight was not the first instance of European authorities flying surveillance missions over the EU's external borders, and the Central Mediterranean in particular. Since 2013, Frontex has awarded a series of contracts to private contractors to build up its own aerial surveillance programme. Yet the introduction of Frontex's MAS service in 2017 marks the beginning of a new phase. Whereas border control in the area had previously been predominantly conducted through naval means, aerial patrols have since come to play a primary role. Without these airborne surveillance operations, the Libyan and Tunisian authorities would not be able to intercept migrants at sea and forcibly return them to their shores.

This report identifies aerial bordering as a key development with wide-reaching consequences for the protection of migrants' rights and for the accountability of state and non-state actors implicated in these abuses. It provides a genealogy of this phenomenon, locating its development within the longer history of European authorities' attempts to prevent migrant crossings in the Mediterranean regardless of the human cost. The report further explains why aerial surveillance has become so prominent in European border enforcement and why this constitutes a serious human rights issue. In this sense, the report serves as a critical companion to the database section of this platform, allowing readers to contextualise and make sense of the data-driven analysis provided therein.

Chapter 1 examines the interconnected dynamics that underscore the shift in European border control from maritime operations to aerial surveillance—that is, from sea to sky. It is guided by a simple question: what is the problem that aerial surveillance seeks to solve?

Part one charts the retreat of European naval assets and the subsequent near abandonment of search and rescue operations at sea. This retreat was predicated on the simultaneous expansion of Libyan and Tunisian actors' presence at sea, initiated and made possible through European support. Part two recalls the key developments in this process of "border externalisation" and its disastrous effects on migrants. Part three traces the extraordinary rise of the aerial surveillance regime up to the introduction of MAS and outlines the steps Frontex undertook to establish its systematic aerial presence. Part four examines the targeting and criminalisation of non-governmental aerial monitoring operations.

Overall, the chapter shows that aerial surveillance allows Frontex to closely monitor the Central Mediterranean while avoiding legal obligations to rescue migrants' boats and disembark them to safety in Europe.

Chapter 2 outlines the main characteristics of the current aerial bordering regime since the introduction of Frontex's MAS service in 2017.

Part one explains how the expansion of Frontex's aerial fleet, through the leasing of manned and unmanned aircraft from private companies, has resulted in a consistent, systematic aerial presence over the Central Mediterranean. It also examines the implications of the ability of Frontex's aircraft to live-stream video and other information to its headquarters and to other European and national authorities, especially concerning decision-making structures and responsibility.

Part two focuses on the role of aerial bordering in what we term the "pushback machine". It identifies three recurring operational patterns that highlight how aerial bordering underpins interceptions in the Central Mediterranean: first, practices of (selective) communication by Frontex; second, the way aerial bordering expands the operational reach of the Libyan Coast Guard; and third, the routine witnessing by Frontex aircraft of aggressive interception practices by Libyan and Tunisian actors, and how that does not lead to any reconsideration of Frontex's role in the "pushback machine". For each of these patterns, a series of exemplary case studies is examined, selected amongst those in the Database based on their relevance and the availability of additional information otherwise absent from Frontex's own records.

In the **Conclusion**, we situate European aerial surveillance in the Central Mediterranean within the transformation of bordering practices at a global level. Here, we highlight three interconnected phenomena: the privatisation of border control; its militarisation; and its entanglement with the Israeli military-industrial complex.

While several European authorities are involved in the aerial bordering regime, this report focuses on Frontex's activities. This is not only because the Agency is arguably one of the main actors driving the growing prominence of aerial bordering in the European context, but also because, despite the increasing opacity shrouding its aerial activities, data concerning the activities at sea of the Agency is comparatively more than that of its military counterparts such as national border guards and the European Union Naval Force Mediterranean (EUNAVFOR Med) missions. Further research is needed to examine the role of these other actors.

Chapter 1—The Shift from Sea to Sky (2013–2017)

Until recently, the surveillance of the EU's external borders in the Central Mediterranean was predominantly carried out at sea. Although planes have since long been used as a tool to monitor large swathes of the Mediterranean, their use was relatively sporadic. Instead, maritime vessels were the chosen means of surveillance, enabling European authorities to be physically present over extended periods of time in areas crossed by migrants' boats and intervene directly by stopping vessels deemed suspicious, destroying empty boats, or deterring so-called "smugglers".

European authorities have thus always had a strategic interest in maintaining a physical presence at sea. This presence, however, entails a series of legal obligations.[1] Most notable among these, as enshrined in maritime law, is the duty to rescue people in distress. Moreover, once taken onboard, rescued passengers are protected by various other bodies of law, which oblige ships to disembark rescued migrants in a place of safety, while prohibiting their return to places where their human rights are routinely violated, such as Libya and Tunisia—popular points of departure for migrants. Despite sustained attempts by European authorities to evade these responsibilities, they could not completely ignore them. This created an impasse: how to monitor developments at sea, without becoming implicated in rescue operations resulting in migrants' safe disembarkation in Europe—an outcome that European politicians have consistently sought to avoid.

The solution was found by shifting surveillance from sea to sky, where such obligations do not apply—or at least not in the same way. While an aircraft pilot remains obliged to notify the relevant authorities of situations of distress at sea, aircraft are limited in their ability to stand by for prolonged periods, let alone embark people in distress.[2] One thing they can do very efficiently, though, is to spot migrant boats at sea, and inform other ships of their location, in this case those of Libyan and Tunisian authorities, so that the latter can forcibly stop them and return them to their place of departure.

A split along a vertical axis thus ensued: at sea, European authorities have withdrawn much of their naval means, while simultaneously emboldening the presence of their Libyan and Tunisian counterparts to serve as proxies; while in the air, the presence of European aircraft has soared, enabling the early detection of migrant boats, without which Libyan and Tunisian ships would not be able to intercept them. This chapter unpacks this vertical split, locating this synchronous double movement—the retreat of European assets at sea and the expansion in the sky—within the longer history of European efforts to stop migrants' crossings in the Central Mediterranean.

1 The legal landscape regulating search and rescue activities at sea, the disembarkation of rescued passengers, and the treatment of asylum seekers is vast and complex and is not the main subject of this report. European Parliamentary Research Service, "Responsibility for Search and Rescue of Migrants in the Mediterranean," PE 762467, October 2024, https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2024)762467.

2 International Maritime Organization and International Civil Aviation Organization, *International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual*, vol. III: Mobile Facilities, 2025 ed.

1. The Retreat of European Authorities at Sea

Border control and migrant rescue activities in the Central Mediterranean have changed profoundly over the past few years, with operational zones expanding and contracting and the number and type of naval assets periodically increasing and decreasing. Despite these ebbs and flows, a broader perspective reveals a clear overarching trend of European authorities' maritime retreat. This retreat has taken place in parallel with active attempts to expel non-governmental organisations that, since 2015, have deployed their own search-and-rescue ships to the Central Mediterranean.

In this section, we argue that, while complex and multi-faceted, this story can be effectively summarised as European authorities' continuous effort to physically remove themselves and European non-governmental actors from areas at sea where they are most likely to encounter migrant boats. As these dynamics have been the subject of extensive scrutiny by activists, journalists, academics and human rights organisations, they are summarised only briefly here.



From *Mare Nostrum* to *Triton*

The military-humanitarian Mare Nostrum operation, a year-long naval operation focused on rescuing migrants at sea, was launched by the Italian government in the aftermath of the infamous migrant shipwreck of 3 October 2013, in which over 360 people who had departed from Libya lost their lives just a few hundred meters from Italian shores. This shipwreck, followed by an equally tragic one just a week later,[3] called into question what has been referred to as the "left-to-die" tactics of European authorities.[4] These strategies, which oscillated between minimal assistance and outright abandonment of migrant boats at sea, had characterised the European response to the surge in migrant crossings that followed the fall of the Tunisian and Libyan regimes in 2011.

Mare Nostrum, despite its problematic "anti-smuggling" rhetoric,[5] constituted a clear shift away from these tactics.[6] From the summer of 2014 onwards, the operation became the target of criticism from various political figures over its costs and for allegedly constituting a "pull factor" encouraging more migrant crossings. Although these accusations were subsequently disproven,[7] Mare Nostrum was terminated in December 2014.

3 Federica Marsi, "For Survivors of Italy's 2013 Migrant Drowning Tragedy, Justice Is a Long Time Coming," *The New Arab*, February 4, 2021, https://www.newarab.com/features/seeking-justice-italys-2013-migrant-drowning-tragedy.

4 Charles Heller and Lorenzo Pezzani, "The Left-to-Die Boat," Forensic Architecture, 2012, https://forensic-architecture.org/investigation/the-left-to-die-boat.

5 See for instance: Martina Tazzioli, "The Desultory Politics of Mobility and the Humanitarian-Military Border in the Mediterranean: Mare Nostrum Beyond the Sea," *REMHU: Revista Interdisciplinar da Mobilidade Humana* 23, no. 44 (2015): 61–82, https://remhu.csem.org.br/index.php/remhu/article/view/503. See also Maurice Stierl, "A Fleet of Mediterranean Border Humanitarians," *Antipode* 50, no. 3 (2018): 704–24, https://doi.org/10.1111/anti.12320.

6 Its rotating fleet comprised one 135-metre-long amphibious vessel, two frigates, two corvettes, four helicopters, three planes and unmanned aerial vehicles. See: "Mare Nostrum Operation," *Marina Militare*, https://www.marina.difesa.it/EN/operations/Pagine/MareNostrum.aspx. In its most intense phase between January and October 2014, more than 156,000 people were rescued and disembarked to Italy.

7 Charles Heller and Lorenzo Pezzani, *Death by Rescue: The Lethal Effects of the EU's Policies of Non-Assistance*, Forensic Oceanography (affiliated with Forensic Architecture, Goldsmiths, University of London), June 2017, https://content.forensic-architecture.org/wp-content/uploads/2023/04/2016_Report_Death-By-Rescue.pdf.

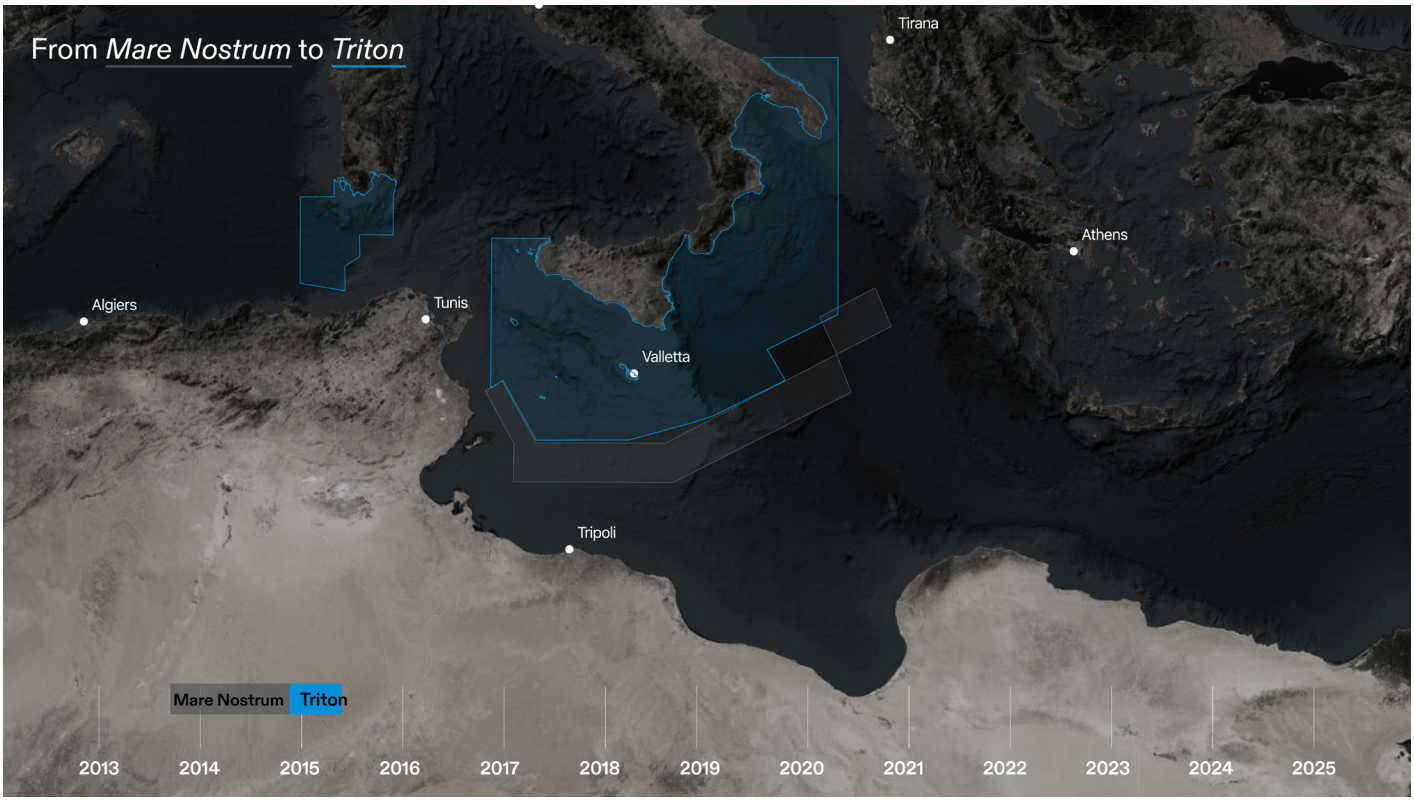


Figure 1.1: Map showing the spatial shift from Italian Operation Mare Nostrum to Frontex' Operation Triton

On 1 November 2014, in response to the planned termination of Mare Nostrum, Frontex, the European Border and Coast Guard agency, launched operation Triton. In contrast to Mare Nostrum, Triton prioritised border surveillance, not rescue at sea; deploying fewer vessels further away from the Libyan coast.[8] As later demonstrated,[9] this retreat formed part of a strategy to deliberately create a "rescue gap" to deter migrants from embarking on the crossing.

Despite this reduced naval presence, Frontex assets were still obliged to respond to Italian Coast Guard requests to carry out rescue operations closer to the Libyan coast. In a confidential letter to the Italian authorities later leaked to the press, the Agency complained about this situation,[10] suggesting that Frontex viewed the very presence of its ships south of its limited operational area as a potential liability as it exposed them to the rescue obligations enshrined in maritime law.

Frontex retreat: from *Triton expanded* to *Themis*

A shipwreck on 18 April 2015, in which more than 1200 people are believed to have died off the coast of Libya, marked another shift in European authorities' activities at sea: it forced the EU to admit that "It was a serious mistake to bring the Mare Nostrum operation to an end. It cost human lives."[11] and to pledge to extend its maritime presence. In May 2015, Triton's budget was expanded, its operational area significantly extended southwards (up to 138 nautical miles

8 Triton's operational area amounted to one-third of that previously covered by Mare Nostrum and was located much further north than the zone where most rescue operations had been carried out under Mare Nostrum.

9 Heller and Pezzani, *Death by Rescue.*

10 Frontex, letter from Klaus Rösler, Director of Operations Division, to Dr. Giovanni Pinto, General Director of Immigration and Border Police, "Concerns about engagement of Frontex deployed assets in activities outside the operational area", Warsaw, 25 November 2014, https://s3.documentcloud.org/documents/3531242/Rosler-Pinto-Frontex-Letter-2014.pdf.

11 Jean-Claude Juncker, "Tackling the Migration Crisis," speech at the European Parliament debate on the conclusions of the Special European Council of 23 April, Strasbourg, April 29, 2015, European Commission, https://ec.europa.eu/commission/presscorner/detail/en/SPEECH_15_4896.

from Italian coasts), and the number of its vessels and aircraft increased.[12] Its main operational task, however, remained border control. Fabrice Leggeri, Frontex's then-director, was quick to declare that: "Triton cannot be a search-and-rescue operation. [...] This is not in Frontex's mandate."[13]

Although Triton played a limited role in rescues, in the summer of 2017 then-Italian Interior Minister Marco Minniti requested a revision of the operation as part of a coordinated plan to stem migrant arrivals, a plan which also included the criminalisation of non-governmental rescue organisations.[14] In February 2018, Triton was replaced by a new Frontex operation, Themis.

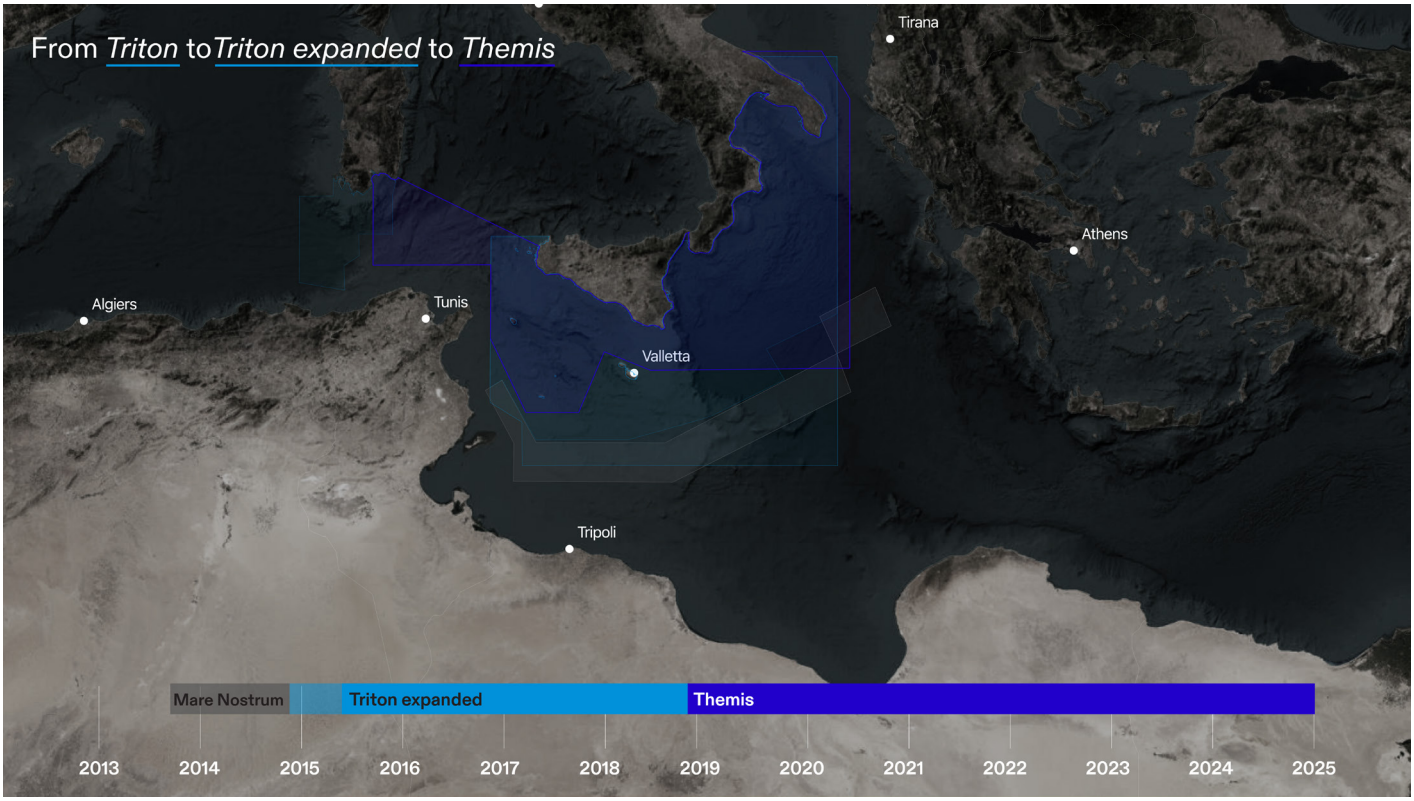


Figure 1.2: Map showing the extend of Frontex Operation Themis

One of the main operational shifts in Themis was the pledge that rescued migrants would not be disembarked by default in Italy, as had happened under Triton. The new mission's operational zone was also decisively shrunk in the Sicily Channel and simultaneously expanded in the Adriatic Sea and south of Sardinia, where rescues have historically been much less frequent. Finally, for the first time in a Frontex mission, Themis included a counter-terrorism security focus, the "collection of intelligence [to detect] foreign fighters and other terrorist threats at the external borders".[15]

Operation Themis remains active to this day, yet the number of migrants rescued by Frontex ships remains marginal. Our data analysis reveals that across the entire period of 2016–2023, Frontex initiated only 4.1% of rescues

12 "Frontex Expands Its Joint Operation Triton," Frontex, May 26, 2015, https://www.frontex.europa.eu/media-centre/news/news-release/frontex-expands-its-joint-operation-triton-udpbHP.

13 Patrick Kingsley and Ian Traynor, "EU Borders Chief Says Saving Migrants' Lives 'Shouldn't Be Priority' for Patrols," *The Guardian*, April 22, 2015, https://www.theguardian.com/world/2015/apr/22/eu-borders-chief-says-saving-migrants-lives-cannot-be-priority-for-patrols.

14 Lorenzo Bagnoli and Andrea Palladino, "Themis: la missione di Frontex voluta da Minniti di cui ora dispone Matteo Salvini," *Open Migration,* June 13, 2018, https://openmigration.org/analisi/themis-la-missione-di-frontex-voluta-da-minniti-di-cui-ora-dispone-matteo-salvini/; Alessandra Giada Dibenedetto, "L'operazione Themis e il suo significato per l'Italia," *Centro Studi Internazionali,* March 27, 2018, https://www.cesi-italia.org/it/articoli/loperazione-themis-e-il-suo-significato-per-litalia.

15 "Frontex launching new operation in Central Med," Frontex, February 1, 2018, https://www.frontex.europa.eu/media-centre/news/news-release/frontex-launching-new-operation-in-central-med-yKqSc7.

leading to disembarkation in Italy or Malta. Even during recent spikes in migrant arrivals to Italy, such as in 2023, Frontex's response has been to agree to step up its operations in the sky (doubling aircraft flight hours) and on land (providing "experts" for the registration and identification of migrants in port cities), while only "considering" an increase in patrolling hours for its vessels at sea.[16]

EUNAVFOR Med: from *Sophia* to *Dry Sophia* and *Irini*

In June 2015, in response to the 18 April shipwreck, the EU launched the anti-smuggling mission of the European Union Naval Force Mediterranean (EUNAVFOR Med), dubbed "Operation Sophia". In line with European authorities' rhetoric against smugglers, conveniently singled out as the culprits of migrant deaths at sea, the Operation's mandate focused on anti-smuggling activities, despite the prominence given to rescue activities in public communications.[17] Spatially, Operation Sophia represented the first significant southward expansion of European presence at sea since Mare Nostrum and aimed to disrupt (migrant) smuggling networks.[18]

While its involvement in rescue operations remained limited compared to other actors,[19] in absolute terms the number of people assisted by Sophia's ships was still significant, with "close to 40000 people" rescued until July 2017. [20] This exposed Sophia to the same critiques that Mare Nostrum had attracted, namely that the operation constituted a "pull factor" leading to more migrants attempting to reach Europe.

This criticism led to two revisions of Sophia's operational plans,[21] including the Italian government demanding that rescued migrants no longer be brought to Italian ports by default in July 2017.[22] Within this tense political context, a key turning point was the "closed ports" policy introduced by the Italian Minister of the Interior Matteo Salvini, who, immediately after taking office in June 2018, began to deny the disembarkation of rescued migrants in Italian ports so as to

16 "Frontex boosts support to Italy," Frontex, September 20, 2023, https://www.frontex.europa.eu/media-centre/news/news-release/frontex-boosts-support-to-italy-IHEK3y.

17 On Sophia's mandate: "About Us," EUNAVFOR Operation Sophia, September 24, 2015, https://web.archive.org/web/20250216011342/https://www.operationsophia.eu/about-us/. This mandate was then reviewed on 20 June 2016 and one year later on 25 July 2017: "EUNAVFOR MED Operation Sophia: mandate extended until 31 December 2018, "European Council, July 25, 2017, https://www.consilium.europa.eu/en/press/press-releases/2017/07/25/eunavformed-sophia-mandate-extended/. On Sophia's limited SAR activities: Eugenio Cusumano, "Migrant rescue as organized hypocrisy: EU maritime missions offshore Libya between humanitarianism and border control," *Cooperation and Conflict* 54, no. 1 (2018): 3-24, https://doi.org/10.1177/0010836718780175.

18 Not only did Sophia's operational zone extend up until the limit of Libyan territorial waters, but the subsequent phases of its mandate—which in the absence of a UN mandate or Libyan request never materialised—foresaw its expansion into Libyan territory, at sea and even on land.

19 Operation Sophia contributed to 13% of rescues executed. Its assets allegedly did not reveal their positions during distress cases so as not to participate in rescue operations. See: Eugenio Cusumano, "Migrant Rescue as Organised Hypocrisy: EU Maritime Mission Offshore Libya between Humanitarianism and Border Control," Cooperation and Conflict 54, no. 1 (2019): 3–24, https://doi.org/10.1177/0010836718780175.

20 "EUNAVFOR MED Operation Sophia: mandate extended until 31 December 2018," European Council, July 25, 2017, https://www.consilium.europa.eu/en/press/press-releases/2017/07/25/eunavformed-sophia-mandate-extended/.

21 Monitoring of the Libyan Coast Guard training; surveillance of oil exports from Libya; and enhanced information sharing with European police forces. See: "EUNAVFOR MED Operation Sophia: mandate extended until 31 December 2018," European Council, July 25, 2017, https://www.consilium.europa.eu/en/press/press-releases/2017/07/25/eunavformed-sophia-mandate-extended/

22 Angelino Alfano, Minister of Foreign Affairs and International Cooperation, statement during *Audizione sulle comunicazioni del Governo sull'evoluzione della situazione in Libia*, Commissioni riunite Affari Esteri e Comunitari (III) e Difesa (IV) della Camera dei deputati e Affari Esteri, Emigrazione (3ª) e Difesa (4ª) del Senato della Repubblica, Resoconto stenografico, session of 1 August 2017. leg.17.stencomm.data20170801.U1.com0304c0304.audiz2.audizione.0028.pdf

impose their relocation to other EU member states. To avert the looming possibility of a long standoff with Italian authorities and of "forc[ing] [Libyan authorities] to become the primary actor [...] in their area of responsibility", Sophia's ships retreated to a "second line" away from Libyan coasts,[23] and its rescue activities effectively ceased.[24] In March 2019, all of Sophia's maritime assets were withdrawn,[25] leading to the paradox of a "naval mission without any ships".[26]

One year later, the EU replaced the operation, reportedly referred to by EU policymakers as 'Dry Sophia' during its final year,[27] with Operation Irini. The decision, triggered by the deteriorating security situation in Libya, involved the redeployment of EU naval assets to the Mediterranean, but in a manner explicitly designed to minimise the "risk" of rescuing migrants.[28] Irini's main objective was the implementation of the UN arms embargo on Libya, with the disruption of migrant smuggling networks as a secondary task to be pursued exclusively "through [...] patrolling carried out by aerial assets above the high seas."[29] Vessels were deployed only to the north and the east of areas typically crossed by migrant boats,[30] and the operation included a monitoring mechanism which could trigger a "repositioning clause", removing vessels from the sea if they were found to generate a "pull factor".[31] These operational modalities show how the very presence of ships at sea—even when those ships were deliberately avoiding rescue involvement—was perceived by many European policymakers as a liability to be avoided.

23 Council of the European Union and European External Action Service, Political and Security Committee, *EUNAVFOR MED Op SOPHIA — Six Monthly Report, 1 December 2018–31 May 2019*, EEAS (2019) 763, Brussels, July 2019, 4, https://s3.documentcloud.org/documents/6294327/EUNAVFOR-MED-Op-SOPHIA-Six-Monthly-Report.pdf

24 Forensic Oceanography, "The Nivin Case: Migrants' Resistance to Italy's Strategy of Privatized Push-Back," Forensic Oceanography (affiliated with Forensic Architecture, Goldsmiths, University of London), December 2019, 25, https://content.forensic-architecture.org/wp-content/uploads/2019/12/2019-12-18-FO-Nivin-Report.pdf.

25 "EUNAVFOR MED Operation Sophia: mandate extended until 30 September 2019," European Council, March 29, 2019, https://www.consilium.europa.eu/en/press/press-releases/2019/03/29/eunavfor-med-operation-sophia-mandate-extended-until-30-september-2019/

26 Silvia Blanco, "Sophia: The EU naval mission without any ships," *El País, September 2, 2019,* https://english.elpais.com/elpais/2019/08/29/inenglish/1567088519_215547.html

27 Federico Petrangeli, "Nota Breve: Da Mare nostrum a Sophia a Mediterraneo sicuro: dieci anni di operazioni navali nel Mediterraneo centrale," *Servizio Studi del Senato,* April, 2023, https://www.senato.it/service/PDF/PDFServer/BGT/01373655.pdf.

28 This was explicitly acknowledged by the Italian Navy, see: "EUNAVFOR MED Operation Irini," Marina Militare, https://www.marina.difesa.it/cosa-facciamo/per-la-difesa-sicurezza/operazioni-in-corso/Pagine/eunavformed-irini.aspx; Jacopo Barigazzi, "EU launches naval mission to police Libya arms embargo," *Politico,* March 26, 2020*,* https://www.politico.eu/article/eu-launches-naval-mission-to-police-libya-arms-embargo/.

29 Council of the European Union, *Council Decision (CFSP) 2020/472 of 31 March 2020 on a European Union military operation in the Mediterranean (EUNAVFOR MED IRINI)*, Official Journal of the European Union, L 101/4, http://data.europa.eu/eli/dec/2020/472/oj.

30 Barigazzi, "EU launches naval mission to police Libya arms embargo."

31 EUNAVFOR MED Operation Irini," Marina Militare, https://www.marina.difesa.it/cosa-facciamo/per-la-difesa-sicurezza/operazioni-in-corso/Pagine/eunavformed-irini.aspx

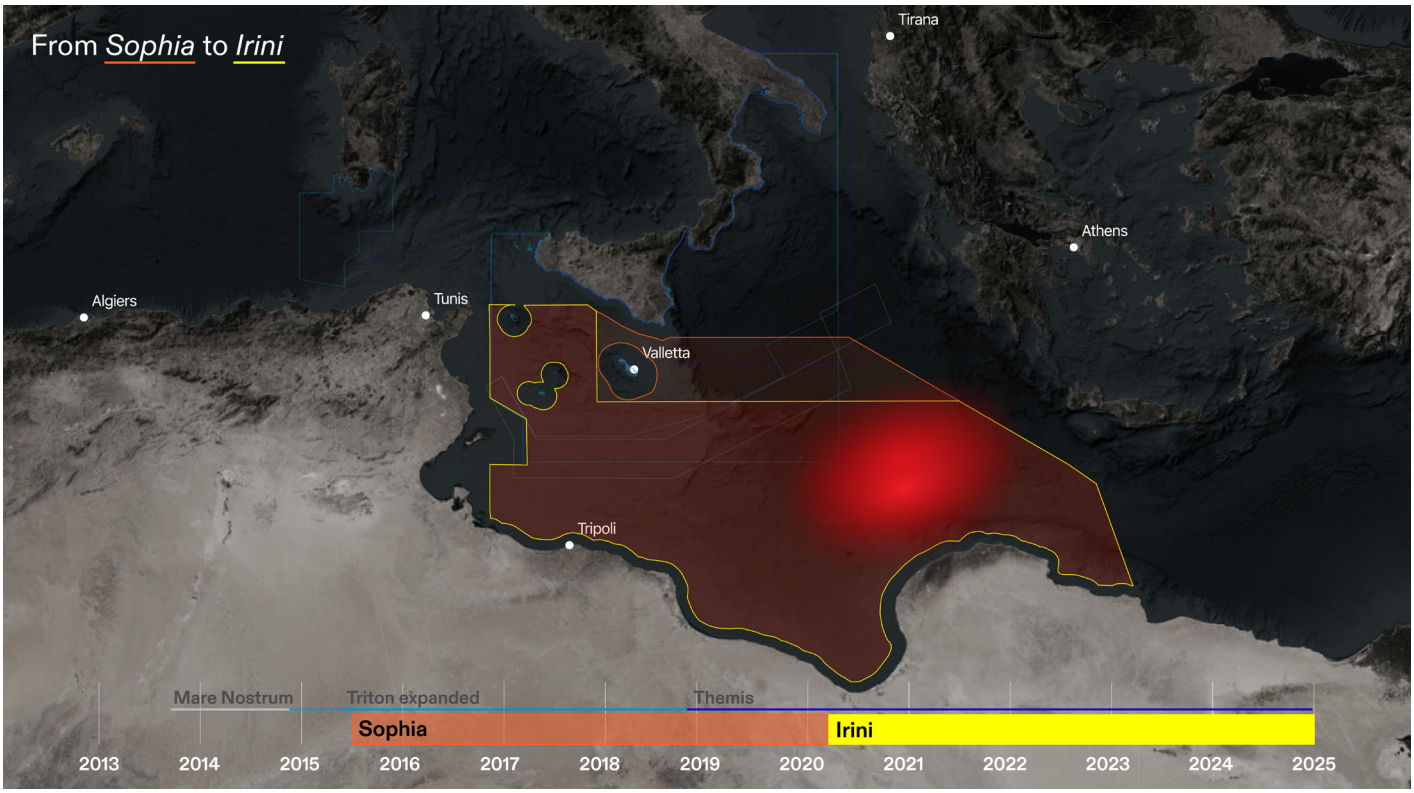


Figure 1.3: Map showing the transition from EUNAVFOR Med Operation Sophia to Irini, with a highlight on the latter's operational focus on the eastern part of the Central Mediterranean

Expelling non-governmental organisations from the sea

This deliberate process of retreat has gone hand in hand with efforts to expel from the sea the many civil society organisations that, since 2015, have attempted to fill the "rescue gap" left by European authorities by deploying their own ships in the Central Mediterranean. More than 20 non-governmental vessels have taken part in rescue missions in 2025, and several others have been part of this "civil fleet" over the years,[32] collectively contributing to the rescue of 175,595 people over a decade of operations.[33]

Since late 2016, maritime rescue NGOs have been the target of a sustained campaign of criminalisation orchestrated by several political groups in Italy and beyond. This campaign has attempted not only to smear their reputation by labelling them smugglers, traffickers, and even pirates, and delegitimise their operations through legal action against their crews, but also to effectively drive them out of the sea through administrative and judicial restrictions targeting the most vital asset of every rescue operation: their vessels.[34]

In Italy, the imposition in 2017 of a so-called "Code of Conduct" by then-Interior Minister Marco Minniti, which aimed to severely restrict NGO activity at sea, was followed a year later by Matteo Salvini's "closed ports" policy. More recently, the 2023 "Decreto Piantedosi" [Piantedosi Decree] has further intensified attempts to restrict and block civilian search-and-rescue activities

32 "What Is the Civil Fleet?", Mediterranea Saving Humans, https://mediterranearescue.org/en/civil-fleet; "Mapping Civil Fleet 2025," Civil MRCC, June 16, 2025, https://civilmrcc.eu/map/mapping-civil-fleet-2025/.

33 "175,000 Lives Saved—A Decade of Civil Search and Rescue in the Central Mediterranean," Sea-Eye, June 18, 2025, https://sea-eye.org/en/175000-lives-saved-a-decade-of-civil-search-and-rescue-in-the-central-mediterranean/.

34 Charles Heller and Lorenzo Pezzani, *Blaming the Rescuers: Criminalising Solidarity, Re-enforcing Deterrence*, Forensic Oceanography (affiliated with Forensic Architecture, Goldsmiths, University of London), June 2017, https://content.forensic-architecture.org/wp-content/uploads/2023/04/2017_Report_Blaming-The-Rescuers.pdf.

in the Central Mediterranean. This, combined with the practice of assigning NGO ships wishing to disembark rescued migrants to ports far away from the area where rescues take place, has forced them into lengthy detours away from their area of operations.

Overall, between August 2017—when the first NGO rescue ship, the *Iuventa, was impounded in Trapani, Sicily for allegedly participating in human trafficking—and June 2025, European countries (Germany, Italy, Malta, the Netherlands and Spain) have initiated a total of 86 administrative or criminal proceedings against civil search-and-rescue actors.*[35] Although all crews have so far been acquitted of all charges,[36] demonstrating their unfoundedness, these proceedings have resulted in a significant drain of NGO resources, a breakdown in cooperation among civil society actors, and a noticeable absence of rescue vessels at sea. Between 2023 and 2024 alone, the Piantedosi Decree has led to a cumulative total of 640 days of detention of NGO ships, as well as an additional 735 days spent at sea reaching disembarkation ports deliberately assigned far from rescue areas.[37]

In parallel, NGO vessels have been exposed to increasing violence by the Libyan Coast Guard, including threats, harassment, dangerous manoeuvres and outright attacks. Between 2016 and 2025, 32 such episodes of violence by Libyan actors against NGO vessels, as well as 24 against migrants, three against Italian fishing boats and one against an Italian Coast Guard ship have been documented by the NGO Sea-Watch.[38] Through these tactics, Libyan actors have directly contributed to efforts to restrict active NGO presence at sea.
Taken together, these elements demonstrate that European authorities' naval retreat was concurrent with their attempts to expel NGOs from the sea.

2. The Externalisation of Naval Operations to Libyan and Tunisian actors

In parallel with the processes of maritime retreat and expulsion outlined above, European authorities have simultaneously worked to embolden Libyan and Tunisian actors' presence at sea. They have done so with a clear mandate: to stop migrant boats and return their passengers to their point of departure. This process of "border externalisation", or the outsourcing of border control, allows European authorities to physically prevent migrants from reaching European shores through their proxies, thus circumventing obligations imposed on them by international and national law.

This section provides a brief overview of some of the most significant steps towards the creation of this arrangement which has involved both the provision of **logistical and operational support in the form of vessels, equipment and training**, and the **establishment of a regulatory framework designed to legitimise Libyan and Tunisian interventions at sea**. As denounced by many human rights organisations and search and rescue groups, these combined measures have enabled Italy and other European authorities to institute a regime

35 European Union Agency for Fundamental Rights, *Search and Rescue Operations in the Mediterranean and Fundamental Rights—June 2025 Update, June 20, 2025,* https://fra.europa.eu/en/publication/2025/june-2025-update-ngo-ships-sar-activities.

36 "Context," Justice Fleet, https://justice-fleet.org/#context.

37 European Union Agency for Fundamental Rights, *Search and Rescue Operations in the Mediterranean and Fundamental Rights — June 2025 Update*, June 20, 2025.

38 "60 Libyan Attacks at Sea as EU Rolls Out Red Carpet for Militias, New Data Shows," Sea-Watch, October 14, 2025, https://sea-watch.org/en/60-libyan-attacks-at-sea-as-eu-rolls-out-red-carpet-for-militias/.

of “refoulement by proxy”, in which they outsource violence and human rights violations against migrants at sea, while maintaining strategic control from the sky.[39]

The process of border externalisation has been the subject of considerable scrutiny and criticism. Rather than offering an exhaustive account, this section provides a concise overview, focusing specifically on its role in relation to aerial bordering within what we define below as the “pushback machine.”[40]

The provision of vessels, equipment and training

Externalisation arrangements in the Central Mediterranean initially focused on (re-)building the logistical and operational capacity of Libyan and Tunisian maritime actors, enabling them to serve as naval proxies for the steadily retreating European fleets.

European authorities focused first on Libya, signing cooperation agreements aiming to stem migrant crossings as early as the late 1990s.[41] In December 2007, Italy pledged to provide the Libyan coast guard with six patrol boats, delivered in May 2009 in the context of an agreement between the two states to conduct joint maritime patrols. This growing collaboration, which was prioritised by European authorities, was interrupted by the fall of the Qaddafi regime in 2011 and the ensuing civil war, which also led to the destruction or damaging of much of the equipment needed by Libyan authorities to conduct maritime patrols, including some of the Italian vessels.

When cooperation with European authorities resumed in 2016, the Libyan Coast Guard was barely functional and “institutionally weak”,[42] with “limited assets, dysfunctional communication and remote sensing equipment.”[43] After signing a Memorandum of Understanding with Libya in 2017, Italy stepped up its support by providing the capabilities and equipment needed to stop migrants at sea. Tracking the exact number of vessels supplied to Libya is difficult, due to the multiplicity of actors involved and the opacity shrouding some of the arrangements.[44] However, Italian investigative outlet IRPI Media found that by December 2022, Italy had supplied and maintained at least 27 vessels for the Libyan General Administration for Coastal Security (7 vessels between 28 and 35 meters long, and about 20 between 9 and 12 meters), and another 17 for the Coast Guard.[45] At least two additional patrol boats have been donated since.[46] In addition to these vessels and other technical equipment and support, European authorities have been conducting several training programmes for the Libyan Coast Guard and Navy, in particular through the EUNAVFOR Med operation.

39 Charles Heller and Lorenzo Pezzani, *Mare Clausum: Italy and the EU’s Undeclared Operation to Stem Migration Across the Mediterranean,* Forensic Oceanography (affiliated with Forensic Architecture, Goldsmiths, University of London), May 2018, https://content.forensic-architecture.org/wp-content/uploads/2019/05/2018-05-07-FO-Mare-Clausum-full-EN.pdf.

40 Civil MRCC, *From Libya to Tunisia: How the EU Is Extending the Push-Back Regime by Proxy in the Central Mediterranean*, June 2024, https://civilmrcc.eu/criminalisation/from-libya-to-tunisia-how-the-eu-is-extending-the-push-back-regime-by-proxy-in-the-central-mediterranean.

41 Heller and Pezzani, *Mare Clausum*.

42 Heller and Pezzani, Mare Clausum, 33.

43 Heller and Pezzani, Mare Clausum, 33.

44 Of the €61.2 million earmarked for the “Support to Integrated Border and Migration Management in Libya” (SIBMMIL), one of the main European funding sources for migration control in Libya, the investigative outlet IRPI Media has been able to track only 56%. No public information exists on how the remaining 44% has been spent. Fabio Papetti, Paolo Riva and Greta Veresan, “Mediterraneo fuorilegge,” *IRPI Media,* March 17, 2026, https://irpimedia.irpi.eu/thebigwall-migranti-attacchi-guardacoste-libia/.

45 Lorenzo Bagnoli and Fabio Papetti, “Come l’Italia ha costruito le forze marittime della Libia,” *IRPI Media*, October 27, 2022, https://irpimedia.irpi.eu/thebigwall-grande-muro-mediterraneo-italia-libia/.

46 Eleonora Vasques, “EU Delivers New Patrol Boats to Libya Despite Militia Links,” *Euractiv,* June 23, 2023, https://www.euractiv.com/news/eu-delivers-new-patrol-boats-to-libya-despite-militia-links/.

NAVAL ASSETS DONATED BY ITALY & THE EU TO LIBYA

Year	Donor	Model	IT Name / Nr	LIB Name / Nr
2009-10	Italy	Bigliani-class boat		
	Italy	Bigliani-class boat		
	Italy	Bigliani-class boat		
	Italy	V.5000-class patrol boat		
	Italy	V.5000-class patrol boat		
	Italy	V.5000-class patrol boat		
2017	Italy	Bigliani-class boat	G83 Macchi	PB644 Zuwarah
	Italy	Bigliani-class boat	G86 Buonocore	PB648 Ras Al Jadar
	Italy	Bigliani-class boat	G82 Galiano	PB654 Sabratha
	Italy	Bigliani-class boat	G84 Fortuna	PB656 Zawiya
2018	Italy	Corrubia-class fast patrol boats	G.92 Alberti	PB658 Fezzan
	Italy	Corrubia-class fast patrol boats	G.115 Zannotti	PB660 Ubari
2018-19	Italy	CP500-class coastal patrol boat	CP515	
	Italy	CP500-class coastal patrol boat	CP516	
	Italy	CP500-class coastal patrol boat	CP517	
	Italy	CP500-class coastal patrol boat	CP518	
	Italy	CP500-class coastal patrol boat	CP519	
	Italy	CP500-class coastal patrol boat	CP520	
	Italy	CP500-class coastal patrol boat	CP521	
	Italy	CP500-class coastal patrol boat	CP522	
	Italy	CP500-class coastal patrol boat	CP526	
	Italy	CP500-class coastal patrol boat	CP535	
2020	Italy	Patrol Boat	P-300	
	Italy	Patrol Boat	P-301	
2022	Italy	RHIB		
	Italy	RHIB		
2023	EU-funded / delivered by Italy	Class-300		
	EU-funded / delivered by Italy	Class-300		
	EU-funded / delivered by Italy	Class-300		
	EU-funded / delivered by Italy	Corrubia-class fast patrol boats	G.108 Conversano	PB662 Murzuq

Figure 1.4: List of all known vessels donated to Libya by Italy or the EU, Images courtesy of Sea-Watch

Similar arrangements have also been made with Tunisia over the last few years, by a larger number of partners. Since 2011, Italy has supplied and maintained at least 12 patrol boats, as well as other equipment and "50 4-wheelers designed to monitor the coasts."[47] More recently, Germany provided Tunisia with 12 inflatable boats and 27 boat motors, as well as equipment to repair Coast Guard vessels. In 2023, France announced 25 million euros in aid "enabling Tunisia to buy border policing equipment and train border guards."[48] Finally, the EU pledged €150 million towards the "fight against illegal migration" in the context of the Memorandum it signed with Tunisia in 2023.[49]

Legitimising Libyan and Tunisian intervention: rescue as legal fiction

The provision of logistical and operational support to the Libyan and Tunisian authorities was not sufficient on its own, however. For them to operate with impunity, their interventions at sea had to be made to appear legitimate. This objective has been achieved through the creation of a legal fiction, one where their activities at sea could be presented as "rescues", rather than as violent interceptions and forced returns to detention and abuse.

Efforts focused initially on Libya and the creation of a Libyan Search and Rescue (SAR) region, the maritime zone within which a state is responsible for coordinating rescue operations. The process of establishing the SAR region began in 2016, when the Italian Coast Guard was instructed by the EU to establish a Libyan Maritime Rescue Coordination Centre (MRCC), the technical

47 Civil MRCC, *From Libya to Tunisia*.
48 Civil MRCC, *From Libya to Tunisia*.
49 Civil MRCC, *From Libya to Tunisia*.

and institutional infrastructure needed to coordinate rescue at sea, and to support the Libyan authorities in "identifying and declaring their Search and Rescue Region".[50] In 2017, Libyan authorities unilaterally declared a Libyan SAR zone, despite several journalistic investigations revealing that the MRCC was not in operation until at least October 2024.[51]

A similar approach was taken in Tunisia; European authorities have, since 2021, worked to establish a functioning MRCC and to promote the declaration of a Tunisian SAR region.[52] Various initiatives by European authorities towards the creation of a Tunisian MRCC have been documented.[53] This process reached its conclusion in June 2024, when Tunisia officially declared its SAR zone.[54] The declaration creates a legal and operational framework in which Tunisian interceptions of migrants at sea can be framed as rescue, thus shielding the human rights violations associated with them from legal scrutiny.

Over the following years, several actors and groups have challenged the various interconnected elements that sustain the legal fiction of "rescue". The legitimacy of the Libyan SAR region, which presupposes the presence of a functioning MRCC, and whose declaration needs to follow specific institutional steps, has been contested.[55] Most rescue NGOs operating in the Central Mediterranean do not recognise the legitimacy of the Libyan authorities, and have consistently denounced and documented their violations including: systematically endangering and directly abusing migrants during operations at sea;[56] failing to respond to distress calls;[57] being involved in criminal activities, including high-ranking commanders who have been denounced for human trafficking by the UN;[58] and threatening NGO and merchant vessels conducting rescue operations at sea.[59]

Amnesty International and other rights groups have found that the Tunisian

50 International Maritime Organization, Sub-Committee on Navigation, Communications and Search and Rescue, "Further Development of the Provision of Global Maritime SAR Services: Libyan Maritime Rescue Coordination Centre Project," NCSR 5/INF.17, submitted by Italy, December 15, 2017, https://cdn.theoutlawocean.com/media/pdf/further-development-of-the-provision-of-global-maritime-sar-services-december-2017-italian-submission-to-imo.pdf.

51 Papetti, Riva, and Veresan, "Mediterraneo fuorilegge"; Lorenzo Bagnoli and Fabio Papetti, "Cosa può succedere se la Libia coordinerà per davvero i salvataggi in mare," *IRPI Media*, October 21, 2024, https://irpimedia.irpi.eu/libia-coordinamento-salvataggi-in-mare/.

52 Civil MRCC, *From Libya to Tunisia*.

53 Civil MRCC, *From Libya to Tunisia*.

54 Amnesty International, 4 October 2024 "Joint Statement: Tunisia is Not a lace of Safety for People Rescued at Sea", https://www.amnesty.org/en/latest/news/2024/10/joint-statement-tunisia-is-not-a-place-of-safety-for-people-rescued-at-sea/

55 Emilio Drudi, "Esposto all'IMO per demolire l'alibi della zona SAR libica," Tempi Moderni, March 21, 2020, https://tempi-moderni.net/index.php?p=2020/03/31/esposto-allimo-per-demolire-lalibi-della-zona-sar-libica/; "Mediterranean: As the Fiction of a Libyan Search and Rescue Zone Begins to Crumble, EU States Use the Coronavirus Pandemic to Declare Themselves Unsafe," Statewatch, May 4, 2020, https://www.statewatch.org/analyses/2020/mediterranean-as-the-fiction-of-a-libyan-search-and-rescue-zone-begins-to-crumble-eu-states-use-the-coronavirus-pandemic-to-declare-themselves-unsafe/.

56 Benjamin Bathke, "Sea Rescue NGO Accuses Libyan Coast Guard of Aggression as its Ship is Impounded Again," *InfoMigrants,* November 2, 2023, https://www.infomigrants.net/en/post/52980/sea-rescue-ngo-accuses-libyan-coast-guard-of-aggression-as-its-ship-is-impounded-again; "EU: Shifting Rescue to Libya Risks Lives", Human Rights Watch, June 19, 2017, https://www.hrw.org/news/2017/06/19/eu-shifting-rescue-libya-risks-lives.

57 Lorenzo Tondo, "'It's a Day Off': Wiretaps Show Mediterranean Migrants Were Left to Die," *The Guardian,* April 16, 2021, https://www.theguardian.com/world/2021/apr/16/wiretaps-migrant-boats-italy-libya-coastguard-mediterranean.

58 United Nations Security Council Committee Established pursuant to Resolution 1970 (2011) concerning Libya, "Abd Al Rahman Al-Milad," *Narrative Summaries of Reasons for Listing,* https://main.un.org/securitycouncil/en/sanctions/1970/materials/summaries/individual/abd-al-rahman-al-milad.

59 "Libya's Coast Guard Accused of Firing at Humanitarian Ship in Distress Mission," *Africanews*, August 26, 2025, https://www.africanews.com/2025/08/26/libyas-coast-guard-accused-of-firing-at-humanitarian-ship-in-distress-mission/.

coastguard has also repeatedly used dangerous and unlawful tactics during operations that placed individuals at serious risk and led to fatalities.[60] Reported practices include high-speed manoeuvres producing destabilising waves, physical assault with batons, close-range deployment of tear gas, boat collisions and issuing threats and verbal abuse.

Citing widespread and systematic abuse faced by migrants upon return to these countries, numerous civil society groups, human rights organisations and judges in Italy[61] have argued that neither Libya nor Tunisia can be considered a "place of safety", permitting lawful disembarkation following a rescue at sea, nor a "safe third country" to which asylum seekers may lawfully be transferred.[62] Despite this tireless work of documentation and denunciation, European authorities continue to uphold the legal fiction of rescue, demonstrating its importance in what is referred to below as the "pushback machine".[63]

In the context of aerial surveillance, Frontex mobilises the legal framework of rescue at sea to justify its policy of communicating the position of migrant boats to the Libyan authorities, whom it identifies as the responsible rescue authority, and whom it therefore claims an obligation to inform. However, this position wilfully disregards not only the documented evidence of abuse that calls into question whether such operations constitute genuine rescues, but also the broader legal obligations binding on the Agency, including the prohibition on returning persons to places where they face persecution or torture.

Within the "pushback machine", maintaining a veneer of legitimacy around a system documented as producing violence and abuse at scale is what enables Libyan and Tunisian actors to function as European proxies in intercepting people at sea and forcibly returning them to their point of departure.

3. The Rise of Aerial Surveillance

The context of European maritime retreat and the simultaneous expansion of Libyan and Tunisian capabilities at sea outlined above sets the scene for the rise of Frontex's aerial operations. A reduced presence at sea entails a loss of "situational awareness" and the ability to detect migrant boats, particularly in areas close to the North African coast, where migrant boats can more easily be intercepted by Libyan and Tunisian authorities. Instead, by enabling authorities to monitor departing

60 Amnesty International, *Tunisia: "Nobody Hears You When You Scream": Dangerous Shift in Tunisia's Migration Policy*, November 6, 2025, https://www.amnesty.org/en/documents/mde30/0180/2025/en/.

61 "Landmark Ruling Confirms: Libya Is Not a Place of Safety!," Sea-Watch, https://sea-watch.org/en/asso/; Virginia Passalacqua, "The 'Open Arms' Case: Reconciling the Notion of 'Place of Safety' with the Human Rights of Migrants," *EJIL: Talk!,* May 21, 2018, https://www.ejiltalk.org/the-open-arms-case-reconciling-the-notion-of-place-of-safety-with-the-human-rights-of-migrants/; ASGI, "Tribunale di Crotone: Illegittimo il fermo della Humanity 1," July 3, 2024, https://www.asgi.it/allontamento-espulsione/tribunale-di-crotone-illegittimo-il-fermo-della-humanity-1/; "Relief as Italian Court Says Rescued Refugees Cannot Be Sent Back to Libya," *Al Jazeera,* February 27, 2024, https://www.aljazeera.com/news/2024/2/27/relief-as-italian-court-says-rescued-refugees-cannot-be-sent-back-to-libya.

62 UN High Commissioner for Refugees (UNHCR), *UNHCR Position on the Designations of Libya as a Safe Third Country and as a Place of Safety for the Purpose of Disembarkation Following Rescue at Sea*, September 2020, https://www.refworld.org/policy/countrypos/unhcr/2020/123326; "IOM and UNHCR Condemn the Return of Migrants and Refugees to Libya," IOM and UNHCR, June 17, 2021, https://mena.iom.int/news/iom-and-unhcr-condemn-return-migrants-and-refugees-libya; Amnesty International, *Externalization of Migration and the Impact on the Human Rights of Migrants: Submission to the UN Special Rapporteur on the Human Rights of Migrants*, June 11, 2025, https://www.amnesty.org/en/documents/eur01/9485/2025/en/; Human Rights Watch, "EU/Italy/Libya: Disputes over Rescues Put Lives at Risk," July 25, 2018, https://www.hrw.org/news/2018/07/25/eu/italy/libya-disputes-over-rescues-put-lives-risk; Human Rights Watch, "EU: Shifting Rescue to Libya Risks Lives"; Amnesty International, *Tunisia: "Nobody Hears You When You Scream."*

63 "EU External Partners: Libya and Tunisia Continue Crimes and Abuse—EU Continues Funding," ECRE, July 14, 2023, https://ecre.org/eu-external-partners-libya-and-tunisia-continue-crimes-and-abuse-eu-continues-funding/.

boats while at the same time avoiding the legal rescue obligations held towards migrants, aerial surveillance has come to play a fundamental role in operations.

This section charts the rise of aerial bordering from the sporadic flights of the early 2000s to the establishment by Frontex of a dedicated fleet of aircraft operated by private contractors capable of maintaining a consistent presence over the Central Mediterranean. The following section examines the operational and regulatory shifts that made this transition possible.

Building Frontex's aerial fleet: regulatory changes, new private actors and drones

The deployment of aerial assets by European authorities over the Central Mediterranean is not a new phenomenon, but until the introduction of Frontex's Multipurpose Aerial Surveillance (MAS) service in 2017 airplanes and helicopters were deployed primarily on the basis of specific operational needs during targeted search-and-rescue or surveillance activities. As recalled by a now-retired official of the Italian Coast Guard, the agency that was then primarily responsible for aerial activities, in the early 2000s "air surveillance was not so frequent. [...] We planned our operations precisely in relation to actual SAR situations [....], so I cannot speak of a real, organised air surveillance in relation to migratory flows in those years."[64]

With the increase in migrant crossings in the Sicily Channel, the situation started to change. As of December 2004, around the same time Frontex was established, the Italian Coast Guard's Catania Aeronaval Base underwent a significant expansion of infrastructure and fleet, at the time making it one of the most important aerial actors in the Central Mediterranean.[65]

The further rise in sea crossings that followed the so-called Arab Uprisings in 2011 prompted a wider range of actors to rely on aerial assets in their operations. For Frontex, as well as agencies such as the Italian Guardia di Finanza, the Italian Navy, and later the European Union Naval Force Mediterranean (EUNAVFOR Med), aerial surveillance began to play an increasing role in the management of migration in the Central Mediterranean. Joint Operation Hermes, Frontex's first operation to focus specifically on the Sicily Channel in early 2011, had already deployed a rotating fleet of airplanes, as did its subsequent Joint Operations in the area, Triton and Themis.[66]

During this phase, however, aerial surveillance remained limited. It would take Frontex almost a decade to build its aerial fleet. This development was both a reflection of, and in part driven by, the broader expansion of the Agency's role and activities, during which its budget soared, its tasks multiplied, its cooperation with other EU agencies strengthened, and its personnel expanded.

64 Sandro Gallinelli, former Admiral, Italian Coast Guard, interview by Chiara Denaro, March 19–20, 2024.

65 Initially equipped with medium-range aircraft like the P-166 DL3, the base underwent substantial upgrades to accommodate long-range aircraft as well, such as the ATR 42 and helicopters like the AB-412 and AW-139: "Guardia Costiera—Base Aeromobili ed elicotteri di Catania," *Aviation Reporter*, February 22, 2024, https://www.ab-aviationreporter.com/guardia-costiera-base-aeromobili-ed-elicotteri-di-catania/; Raffaele Fusilli, "2 Nucleo Aereo Guardia Costiera base aeromobili Catania," *Aviation Report*, December 2, 2013, https://www.aviation-report.com/2-nucleo-aereo-guardia-costiera-base-aeromobili-catania/; Sandro Gallinelli, former Admiral, Italian Coast Guard, interview by Chiara Denaro, March 19–20, 2024.

66 The exact number of aircraft deployed is not easy to reconstruct exactly as detailed operational plans are not available to the public and from publicly available information (press releases, commentaries, etc.) it is not clear if the number of assets mentioned refers to overall number of assets deployed over the course of the operation on a rotating basis or concurrently. See for instance: "Frontex Expands Its Joint Operation Triton," Frontex, May 26, 2015, https://www.frontex.europa.eu/media-centre/news/news-release/frontex-expands-its-joint-operation-triton-udpbHP.

A series of regulatory shifts and operational tests made the operational debut of Frontex's key programme, Multipurpose Aerial Surveillance (MAS), possible. A turning point was the introduction of a new Frontex regulation in 2011, establishing that "the Agency may acquire [...] or lease technical equipment for external border control to be deployed during joint operations."[67] This proved to be crucial for the implementation of the aerial surveillance programme, as it allowed Frontex for the first time to independently procure its aerial assets without having to rely on the cooperation of EU member states.

After a failed attempt to purchase a surveillance plane to monitor the Greek-Turkey land border in 2013,[68] the following year Frontex contracted the UK-based company Diamond Executive Aviation (DEA) to provide a fixed-wing aircraft for a 40-day period.[69] This enabled the Agency not only to assess the "operational effectiveness" and viability of aerial surveillance, but it also, and perhaps most importantly, allowed it to test new practices for carrying out such services.[70] During this period, the practice of leasing aerial surveillance services from private actors emerged as the "most accessible, qualitative, effective and cost efficient" way of building its aerial fleet, as opposed to the more lengthy and costly option of purchasing its own equipment.[71]

In September 2015, the Agency established its Aerial Surveillance Services (FASS), introducing a framework contract that allowed for the leasing of aircraft for a defined timeframe.[72] This took place shortly before the European Commission, at the height of the so-called "migration crisis," proposed the transformation of Frontex into a fully-fledged European Border and Coast Guard Agency, further expanding its mandate, budget, and operational capacities. As reflected in Frontex's calls for tenders over the years, 2015 thus marks the beginning of the expansion of the Agency's aerial fleet through collaboration with private companies. These have included: Vigilance BV and EASP Air BV (both from the Netherlands), DCI (France), MIKC (Latvia), Diamond Executive Aviation (UK) and its Malta-based sister company ESAO Ltd,[73] Indra Sistemas SA (Spain), and CAE Aviation s.a.r.l. (Luxembourg).[74] This role of private companies will continue to intensify over the following years.

67 European Parliament and Council of the European Union, *Regulation (EU) No 1168/2011 of the European Parliament and of the Council of 25 October 2011 amending Council Regulation (EC) No 2007/2004 establishing a European Agency for the Management of Operational Cooperation at the External Borders of the Member States of the European Union,* https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32011R1168.

68 "EU: Frontex Cancels Surveillance Plane Contract due to Lack of Interest from Companies," *Statewatch*, October 18, 2013, https://www.statewatch.org/news/2013/oct/eu-frontex-cancels-surveillance-plane-contract-due-to-lack-of-interest-from-companies/.

69 While the area of operation of this specific aircraft was not explicitly stated at the time, it was likely the Bulgaria-Turkey border.

70 "EU: Frontex Presses on with Aerial Surveillance Projects," *Statewatch*, August 12, 2014, https://www.statewatch.org/news/2014/aug/eu-frontex-presses-on-with-aerial-surveillance-projects/; "DEA Completes Frontex Pilot Project," Diamond Executive Aviation, July 2014, https://web.archive.org/web/20141109110544/http://www.diamond-executive.aero/showkb.php?org_id=514&kb_id=38169.

71 "EU: Frontex Presses on with Aerial Surveillance Projects," *Statewatch*, August 12, 2014.

72 Frontex, letter from Fabrice Leggeri, Executive Director, to Mr. Matthias Oel, "Written Questions (E-003553/2019, E-003554/2019, E-003555/2019) Submitted by MEP Özlem Demirel," January 14, 2020, https://www.europarl.europa.eu/RegData/questions/reponses_qe/2019/003553/P9_RE(2019)003553(ANN)_XL.pdf.

73 "About Us," European Specialised Airborne Operations, https://esao.aero/about/.

74 After the signature of a first contract in 2015, with Vigilance BV for 275.000 euros, in 2016, Frontex signed two more contracts for the services foresaw by the call for tender "Framework contract for aerial surveillance services assets and expert support" (Frontex/OP/166/2015/JL) with Diamond Executive Aviation (1 458 393,00 euros) and Indra Sistemas SA (139 000,00 euros). This was followed in 2017 by three new contracts[2] with CAE Aviation s.a.r.l (1 746 000,00 euros), Diamond-Executive Aviation (3 929 247,00 euros), and EASP Air BV (2 145 525,00 euros) and in 2018 with Diamond Executive Aviation (4,793,875.00 euros) and EASP AIR BV (1,349,125.00 euros) for the Lot 1, and with DEA for the Lot 2 (999,820.00 euros). Frontex, *Public Register of Documents*, https://prd.frontex.europa.eu/; List of companies retrieved from Matthias Monroy, "Border Drones (Part 1): Unmanned Surveillance of the EU's External Borders by Frontex," *Dissecting Security Architectures,* July 22, 2021, https://digit.site36.net/2021/07/22/border-drones-part-1-unmanned-surveillance-of-the-eus-external-borders-by-frontex/.

Frontex's ability to monitor the EU's maritime borders was also bolstered through collaboration with other EU Agencies such as the European Maritime Safety Agency (EMSA) and the European Fisheries Control Agency (EFCA), with the aim of creating technical synergies in information sharing and surveillance.

In parallel with the use of manned fixed-wing aircraft, Frontex started to deploy unmanned aerial vehicles, more commonly known as drones, several years ago.[75] Drones have been explicitly indicated by the European Commission as a tool to overcome the challenges of detecting migrant boats across the Mediterranean and have been adopted for maritime surveillance operations by numerous other military actors active in the Central Mediterranean.[76]

The introduction of drone operations required a long series of preparatory steps, beginning as early as 2009. In addition to training events, demonstrations by manufacturers and technical assessments, significant regulatory changes were needed to allow drone flights in "non-segregated airspace", i.e., airspace shared by manned and unmanned vehicles.[77]

The first tender for drone testing was issued by Frontex in September 2016 but received no bids.[78] Test flights began in 2018 after a second call resulting in two contracts: one awarded to Israeli Aerospace Industries (IAI) and Airbus for a drone to be stationed in Crete and operate in the Aegean; and the other to the Italian company Leonardo, which deployed its Falco EVO drone from Lampedusa in the Central Mediterranean.[79] A third trial was conducted in Portugal in collaboration with the European Maritime Safety Agency (EMSA), an attempt by the European authorities to involve an agency not primarily dedicated to border enforcement in order to "contribute to a more friendly perception of using this technology".[80]

Together, these preliminary administrative changes and operational tests paved the way for the beginning of a new phase of aerial bordering at the EU's external borders—one in which surveillance from the sky, conducted by a consolidated aerial fleet, plays a central role in the forced return of migrants to Libya and Tunisia.

75 For instance, the Italian police (Ministry of Interior) and border guard agency (Guardia di Finanza), the Italian Navy mission Mare Sicuro, EUNAVFOR Med. Duccio Facchini, "Un drone per sorvegliare il Mediterraneo e fermare i migranti. Il nuovo appalto del ministero dell'Interno," *Altraeconomia*, October 22, 2020, https://altreconomia.it/un-drone-per-sorvegliare-il-mediterraneo-e-fermare-i-migranti-il-nuovo-appalto-del-ministero-dellinterno/; Riccardo Coluccini and Paolo Riva, "I silenzi della guardia di finanza sui droni impiegati nel Mediterraneo," *IRPI Media*, January 29, 2025, https://irpimedia.irpi.eu/sorveglianze-droni-migranti-mediterraneo-guardia-di-finanza/; "Border Surveillance, Drones and Militarisation of the Mediterranean," *Statewatch*, May 6, 2021, https://www.statewatch.org/analyses/2021/border-surveillance-drones-and-militarisation-of-the-mediterranean/.

76 European Commission, Proposal for a Regulation of the European Parliament and of the Council amending Regulation (EC) No 1406/2002 establishing a European Maritime Safety Agency, COM(2015) 667 final, December 15, 2015, https://eur-lex.europa.eu/legal-content/EN-SL/TXT/?uri=CELEX:52015PC0667&from=EN.

77 "Sorveglianza marittima di Frontex con il drone Falco EVO di Leonardo: supporto ENAC e rilascio certificazioni e autorizzazioni per operazioni," Ente Nazionale per l'Aviazione Civile, December 6, 2018, https://comunicati.enac.gov.it/it/announcement/show/sorveglianza-marittima-di-frontex-con-il-drone-falco-evo-di-leonardo-supporto-enac-e-rilascio-certificazioni-e-autorizzazioni-per-operazioni.

78 Monroy, "Border Drones (Part 1)."

79 To assess the drone's surveillance capabilities, as well as its economic and operational efficiency, the EVO was scheduled to complete over 300 flight hours during missions organised by the Guardia di Finanza in collaboration with the Italian Ministry of the Interior. The drone was authorised to operate across both Maltese and Italian airspaces—a development that was relatively unprecedented. See Monroy, "Border Drones (Part 1)."

80 European Commission, COM(2015) 667 final; "RPAS Enter into Operation in Portugal for Border Surveillance," European Maritime Safety Agency, September 26, 2018, http://www.emsa.europa.eu/newsroom/latest-news/item/3366-rpas-enter-into-operation-in-portugal-for-border-surveillance.html.

4. Stopping Civilian Aerial Monitoring

European authorities are not alone in recognising the crucial importance of aerial monitoring at sea. In parallel with the development of Frontex's aerial fleet, the Central Mediterranean has also witnessed the emergence of civilian monitoring operations by rescue NGOs, who have sent out their own aircraft to support non-governmental rescue efforts and document human rights violations committed by the authorities themselves.

Non-governmental aerial operations began in 2017, when the NGO Sea-Watch, in partnership with the Humanitarian Pilots Initiative (HPI), launched its inaugural aerial mission with Moonbird, a small propeller aircraft.[81] In 2018, a second NGO aircraft started its operations: the Colibri, a reconnaissance plane by the Pilotes Volontaires.

Over the following years, the non-governmental aerial fleet expanded significantly. In 2020, both Sea-Watch and Pilotes Volontaires began operating a second aircraft, the *Seabird 1* and the *Colibri 2* respectively. In 2022, the Seabird 2 replaced Sea-Watch's *Moonbird*, while the Seabird 3 was deployed from 2025, in collaboration with HPI and United4Rescue[82]. At the end of 2025, the NGO SOS Méditerranée also deployed an aircraft over the central Mediterranean, the *Albatross*, in partnership with HPI.

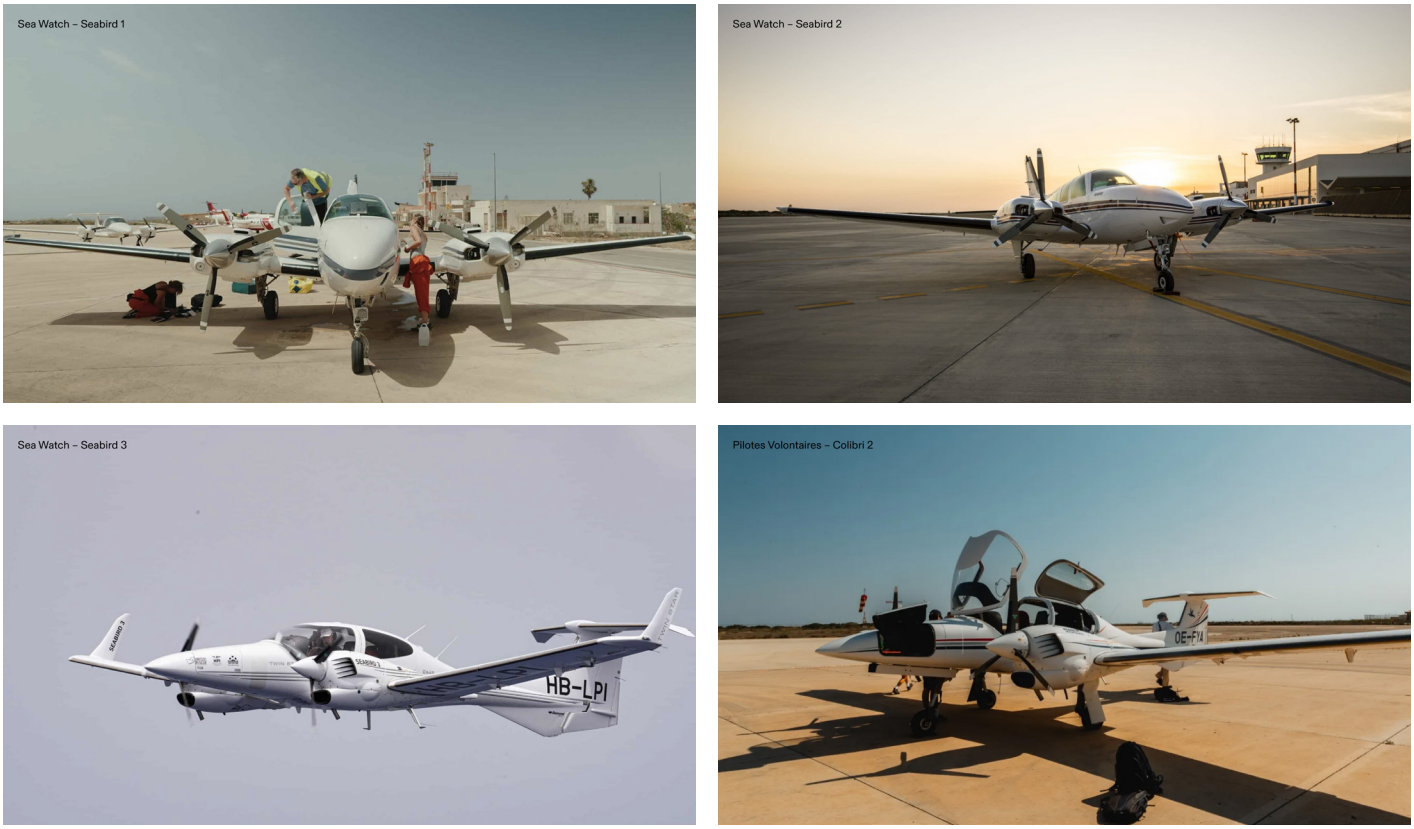


Figure 1.5: Civilian monitoring assets flying in the Central Mediterranean, Images courtesy of Sea-Watch and Pilotes Volontaires

81 The *Moonbird is a Cirrus SR22 plane, capable of flying six-hour missions and covering a maritime area of* 16,000 *km2*.

82 "Seabird 1, 2 & 3," Sea-Watch, https://sea-watch.org/en/mission/airborne/.

Similarly to their maritime counterparts, however, these non-governmental planes have been the subjects of a series of legal and political attacks:

- In 2018, Malta blocked Sea-Watch's *Moonbird* for four months, forcing it to relocate operations to Lampedusa;[83]

- In 2019, Moonbird was prevented from operating from Lampedusa for 3 months by the Italian Civil Aviation Authority (ENAC), which argued that Moonbird did not comply with the requirements for its kind of operations.[84]

- In 2020, ENAC again grounded Moonbird for one month, alleging "spending too many hours at sea and posing a risk of endangering ongoing SAR operations carried out by state actors."[85] That same year, Italy contacted Switzerland, where the HPI is based, to argue the aircraft's operations were commercial and therefore prohibited;[86]

- In 2022, Libyan authorities repeatedly denied Sea-Watch entry to their Flight Information Region, the airspace in which flight information is provided, and blocked flight access to the Libyan SAR zone—an act later confirmed to be unlawful by the German Bundestag Research Service;[87]

- In 2024, ENAC introduced a new decree threatening aircraft detention and issued five orders banning civilian aerial monitoring from key airports in Sicily, Pantelleria, and Lampedusa. Sea-Watch denounced the flight bans as politically driven: "Italy is trying to get rid of the last witnesses to European crimes in the Mediterranean. We will not be intimidated by this;"[88]

- In August 2025, ENAC placed Seabird 1 under administrative detention with immediate effect, resulting in the first seizure of a Sea-Watch civilian monitoring aircraft.[89]

- In September 2025, Pilotes Volontaires' aircraft Colirbi 2 was grounded for 20 days.[90]

Taken together, these bans, flight groundings, and bureaucratic obstacles reveal a clear trend: the criminalisation of search-and-rescue NGOs extends also into the air. By restricting their, European authorities can preserve their monopoly over aerial surveillance, thus making it increasingly difficult for civil society actors to document violations in the Central Mediterranean.

83 Sea-Watch e.V., "Witnesses Not Welcome: Civil Reconnaissance Aircraft Blocked from Operation," July 4, 2018. https://sea-watch.org/en/witnesses-not-welcome-civil-search-aircraft-blocked-from-operation/

84 Felice Florio, "L'Italia ferma il volo degli aerei che segnalano la posizione dei migranti in mare", Open, August 27, 2019, https://www.open.online/2019/08/27/litalia-ferma-il-volo-degli-aerei-che-segnalano-la-posizione-dei-migranti-in-mare/.

85 European Union Agency for Fundamental Rights (FRA), "Search and Rescue Operations in the Central Mediterranean and Fundamental Rights – June 2026 Update," July 28, 2026, Annex, https://fra.europa.eu/en/publication/2026/search-and-rescue-operations-central-mediterranean-and-fundamental-rights-june.

86 Wallis, "Italy Grounds Moonbird Plane."

87 Deutscher Bundestag, Wissenschaftliche Dienste, *Rechtsfragen zum Abkommen über die Internationale Zivilluftfahrt (ICAO-Abkommen)*, WD 2-3000-039/22, June 27, 2022, https://www.bundestag.de/resource/blob/905042/9c57b80506fdddaa3821d4c0a1b9bd0a/WD-2-039-22-pdf-data.pdf.

88 "Italy Bans Human Rights Monitoring over the Mediterranean," Sea-Watch, May 8, 2024, https://sea-watch.org/en/italy-bans-human-rights-monitoring-over-the-mediterranean/.

89 "Italy Detains Human Rights Monitoring Aircraft," Sea-Watch, August 8, 2024, https://sea-watch.org/en/italy-detains-human-rights-monitoring-aircraft/. Pilotes Volontaires, "Our aicraft, Colibri 2 has just been grounded by the italian authorities for 20 days. This detention comes with an outrageous financial penalty of €100,000.," Facebook, September 12, 2025, https://www.facebook.com/pilotesvolontaires/posts/1104382591837616/.

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Chapter 2—Aerial Operations (2017–2025)

Chapter 1 examined how Frontex assembled its vast aerial fleet to patrol the Central Mediterranean. This chapter turns to analysing the Agency's current aerial bordering regime from 2017 onwards.

During this period, Frontex expanded and consolidated its aerial fleet. After 2017, spending on aerial assets leased privately rose sharply, corresponding to the growth of the Agency's dedicated budget. The amount of money the Agency spends on procuring and testing aerial surveillance services from private contractors went from 11.7€ million in 2018 to 278.6 € million in 2026. The last tender published by the Agency on 25 June 2026 amounts to a staggering 522.6€ million over two years in aerial surveillance services. While it is unclear if, when and to whom a contract has been already awarded in this case, it means that, as of October 2026, the Agency has already allocated 330.9€ million for aerial surveillance for 2027 and 220.3€ million for 2028, sums which are set to continue increasing in the next months.

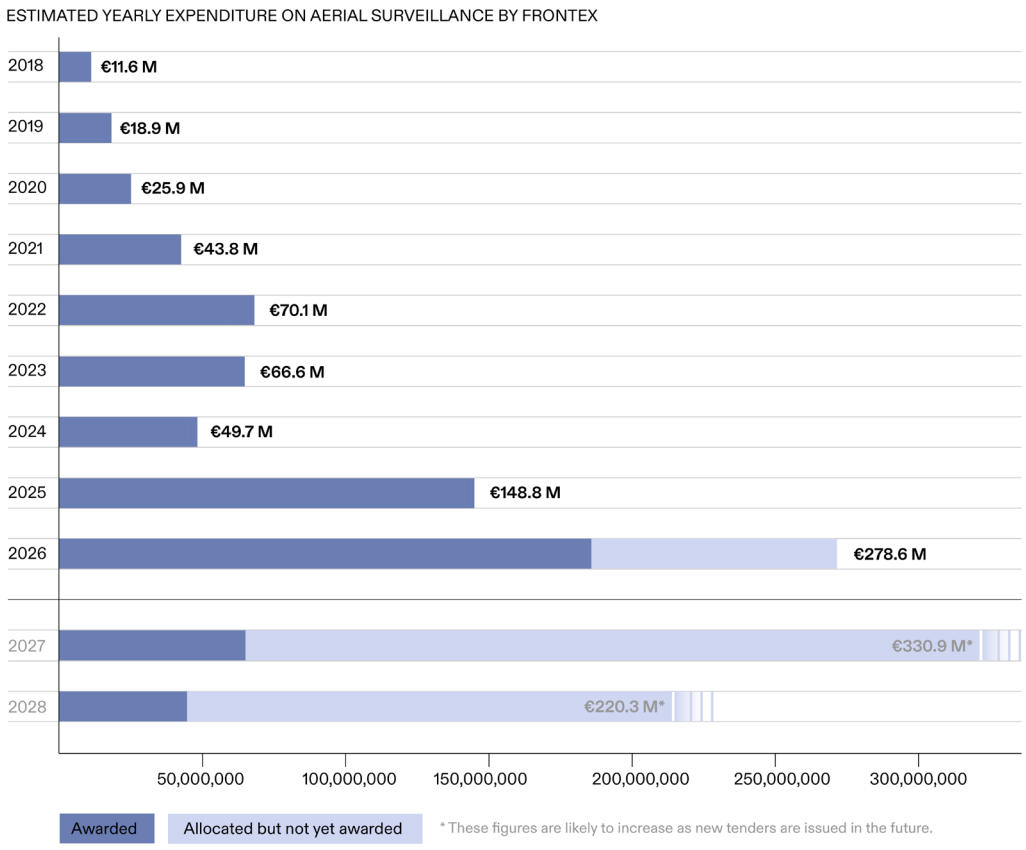


Figure 2.1: Estimated yearly expenditure on aerial surveillance by Frontex.

While the Agency does not publicly reveal the identity of the aircraft operating on its behalf at any given moment in time, thanks in part to the investigative work of journalists and non-governmental groups,[91] LIMINAL was able to compile a list of aerial assets that have been involved in aerial patrols on behalf of the Agency since at least 2019. These include 15 manned aircraft flying out of airports

91 See, for example, Sergio Scandura, X profile, accessed September 2, 2026, https://x.com/scandura; Sea-Watch, "Airborne," accessed September 2, 2026, https://sea-watch.org/en/mission/airborne/.

in Malta, Italy and Greece.[92] In addition, between May 2021 and April 2026, when it crashed at sea, Frontex operated a Heron drone from Malta airport. Manufactured by Israel Aerospace Industries (IAI) and operated by ADAS, a subsidiary of the European aerospace giant Airbus, the drone played a key role in aerial bordering, as described in detail below.

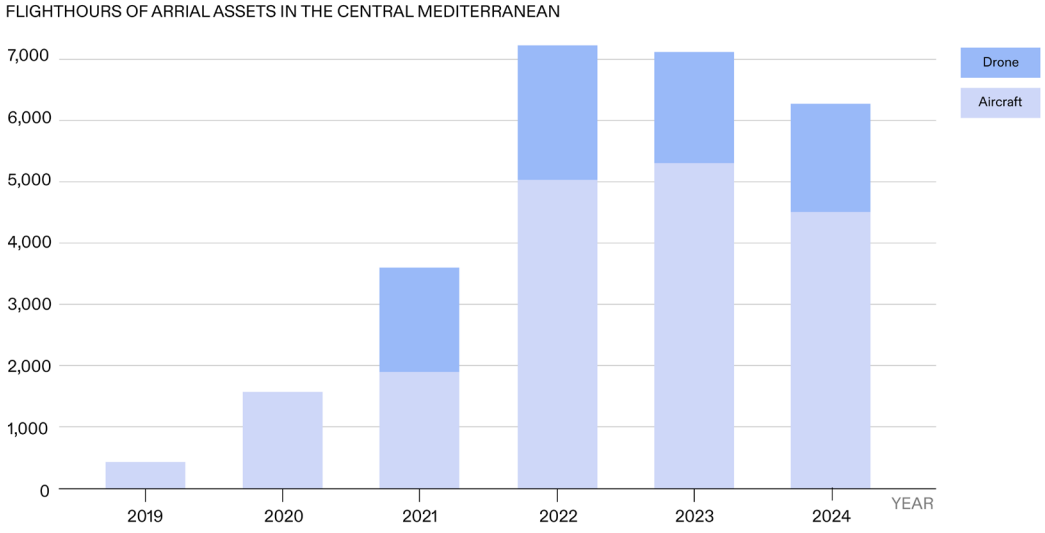


Figure 2.2: Increase of Frontex aerial assets flight hours and Frontex drone flight hours

This expanding fleet marked a turning point. Through it, Frontex transformed the practice of aerial surveillance from the deployment of targeted operations to an extended presence in the sky—what some have referred to in other contexts as a form of "aerial occupation."[93] While this transformation remains an ongoing, continuous process without a clearly defined beginning, we take the introduction of Frontex's Multipurpose Aerial Surveillance (MAS, sometimes referred to as Maritime Aerial Surveillance) service in 2017 to symbolically mark its start.

1. Multipurpose Aerial Surveillance: Towards Aerial Occupation

Frontex describes its Multipurpose Aerial Surveillance service (MAS) as "an integral part of our operations, and a permanent service offered to national authorities. MAS uses surveillance airplanes and drones that stream video and other data from EU and Schengen external borders directly to our headquarters in Warsaw and to national and European authorities, allowing for real-time monitoring."[94] MAS was introduced with the stated aim of improving the "situational awareness" at EU borders and beyond, and delivering "a real-time

92 Frontex also deployed a Hermes 900 drone for border surveillance operations, basing the drone on Crete and conducting flights under the authority of EMSA. In early January 2019, the aircraft crashed on the runway at Tympaki Airport during landing. In July 2022, another IAI Heron drone (the same as the one stationed in Malta) began operating from Crete, carrying out missions over the Ionian Sea. However, the Heron too suffered damage and crashed into the sea 70 nautical miles southeast of Crete. The asset was reportedly on a surveillance flight when it lost satellite communication with its ground station for unknown reasons. See: Matthias Monroy, "Second Frontex Drone Crashed near Crete: Airbus Was Allowed to Fly the Heron 1 alongside Civilian Aircraft," *Dissecting Security Architectures*, August 27, 2023, https://digit.site36.net/2023/08/27/second-frontex-drone-crashed-near-crete-airbus-was-allowed-to-fly-the-heron-1-alongside-civilian-aircraft/.

93 John R. Emery and Daniel R. Brunstetter, "Drones as Aerial Occupation," *Peace Review* 27, no. 4 (2015): 424–31, https://doi.org/10.1080/10402659.2015.1094319. See also: Joshua Hellinger and Ravi Bhavnani, "Aerial Occupation and Aerial Forensics in Gaza," *Global Challenges*, no. 2 (2023), https://doi.org/10.71609/iheid-wrbr-p248. We are grateful to Itamar Mann for pointing us to this concept.

94 "Monitoring and Risk Analysis," Frontex, https://www.frontex.europa.eu/what-we-do/monitoring-and-risk-analysis/monitoring-and-risk-analysis/.

surveillance service" to Member States and EU agencies.[95] Below, we unpack those two characteristics to analyse Frontex's current aerial operations using MAS.

Enhanced situational awareness

Thanks to its vast fleet, Frontex has been able to increase the consistency of its aerial operations, performing daily take-offs from bases in Malta, Lampedusa or Sicily. This has allowed aerial surveillance over the Central Mediterranean to steadily expand, both temporally and spatially.

From a temporal point of view, Frontex flight time off the coast of Libya and Tunisia has grown remarkably, with aircraft having become a near-constant presence over the last few years. Our analysis of flight data gleaned from aircraft tracking websites and other sources demonstrates this steady increase in global flight time for Frontex aircraft. In our Database, this increase is reflected in the darker blue bars representing daily hours flown by Frontex. Drones in particular have played a key role in this temporal expansion. Frontex's introduction of drones for aerial bordering has substantially increased the duration and "endurance" of its aerial missions. Our analysis of the collected flight data for the Central Mediterranean reveals that while a surveillance mission by a manned aircraft typically lasts no more than 5,1 hours, a drone's flight time can reach up to 22,5 hours, with typical missions lasting in average around 11,4 hours.[96]

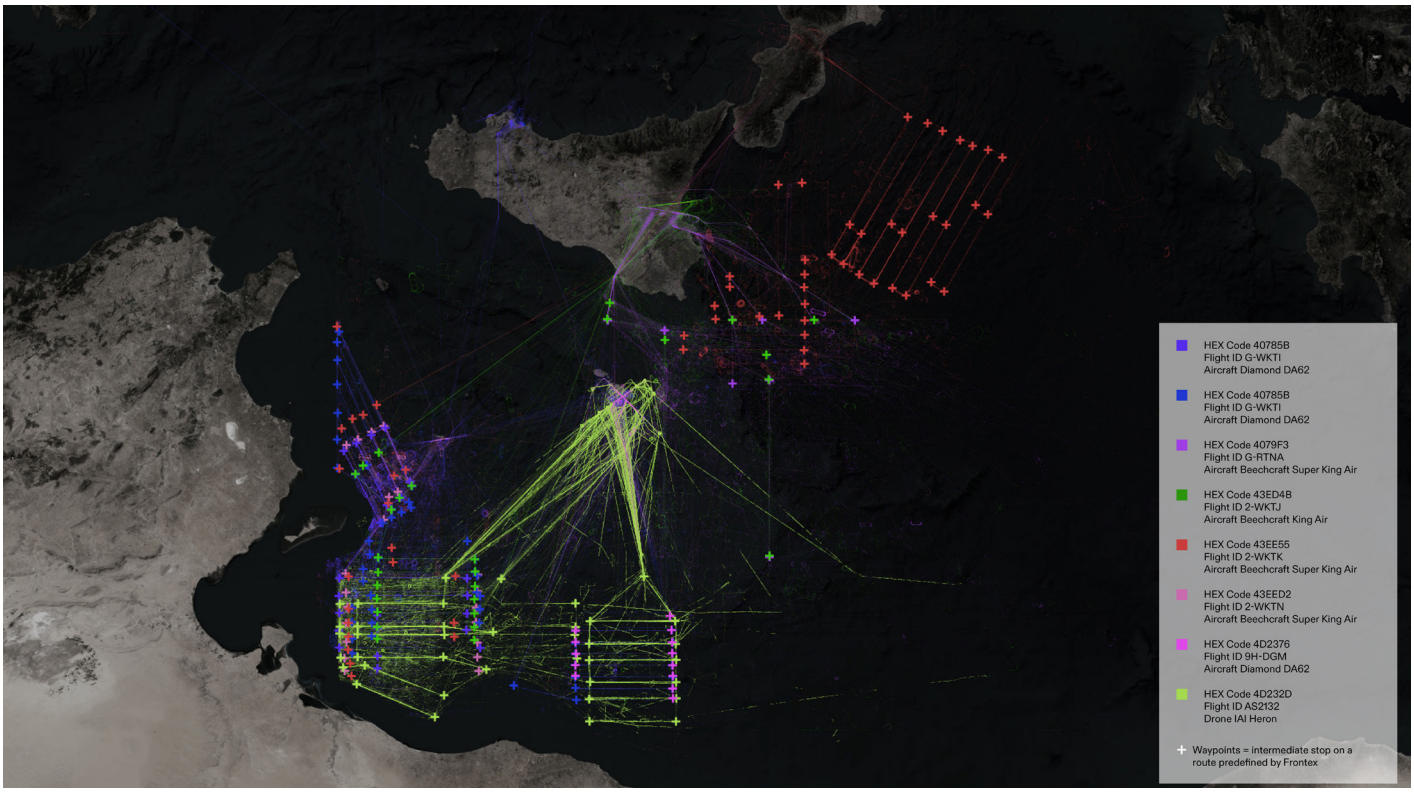


Figure 2.3: All waypoints by Frontex aerial assets combined

95 Frontex, *Multipurpose Aerial Surveillance*, Publications Office of the European Union, 2018, https://op.europa.eu/en/publication-detail/-/publication/b96286e0-1aa8-11e8-ac73-01aa75ed71a1.

96 This operational longevity was also celebrated in PR materials during the drones' testing, when "an extra-long" mission flown by the Falco EVO to support an "interception" lasted 17 hours and 21 minutes. See: "Leonardo's Falco EVO Drone Is Used to Monitor Irregular Migration during Frontex Operation," Leonardo, July 12, 2019, https://www.leonardo.com/en/press-release-detail/-/detail//12-07-2019-leonardo-s-falco-evo-drone-is-used-to-monitor-irregular-migration-during-frontex-operation.

This temporal expansion has also enabled a spatial expansion of the sea areas monitored by Frontex. This means, firstly, that Frontex's aircraft, and the drone in particular, can monitor areas further away from its bases in Malta, Lampedusa or Sicily, and much closer to North African coasts. The Agency itself has acknowledged that it "has never before patrolled so close to Libyan territorial waters".[97] By enabling the detection of migrants' boats early in their journey, this system has made interceptions by Libyan and Tunisian authorities easier. Secondly, this spatial expansion translates into an increase in the extent and the number of areas patrolled each day. While the Agency does not disclose where its assets patrol, publicly available flight tracking data provide important insights into these aircraft's whereabouts. An analysis of flight tracks of both manned planes and the drone over extended periods of time reveals that each aircraft generally follows a standard search pattern over a designated area of the Central Mediterranean, often for several weeks.[98] Together, their daily movements constitute a tightly knit, extensive web of aerial patrol.

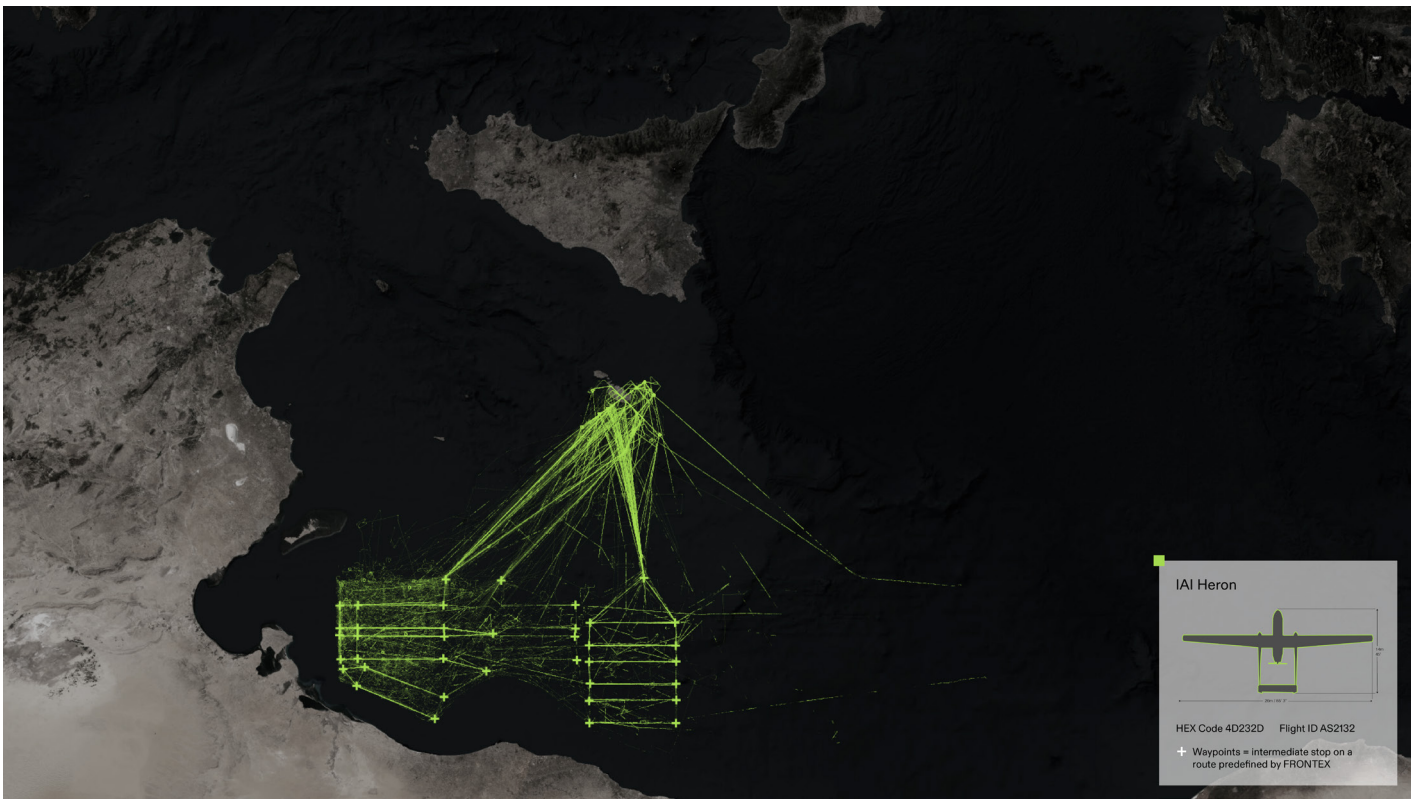


Figure 2.4: Frontex Heron Drone and the waypoints on pre-defined routes

In addition, Frontex continues to develop and operate smaller tactical unmanned assets, whose activities remain beyond public scrutiny as most of the time they are not trackable as their larger counterparts. These include, for instance, vertical-take-off drones operated from patrol vessels that significantly expand the latter's surveillance range.[99]

97 Frontex Press Office, emails from the press office, FOI requested by Arne Semsrott, FragDenStaat, 2021, https://fragdenstaat.de/anfrage/e-mails-der-pressestelle/.

98 The pre-determined pattern resembles what is known as a parallel sweep search; this pattern "is almost always used when a large search area must be divided into sub-areas for assignment to individual search facilities which will be on scene at the same time." *IAMSAR Manual*, vol. II: Mission Co-ordination, chapter 5, 2019 ed., https://cranston.independent-inquiry.uk/wp-content/uploads/cranston-evidence/INQ010511_Extract_from_IAMSAR__International_Aeronautical_and_Maritime_Search_and_Rescue__Manual__Chapter_5_of_Volume_II__2019_Edition.pdf.

99 Frontex, "Frontex and Italy Complete Trial of the Largest Drones the Agency Has Yet Flown from a Patrol Ship," *Frontex, 2026,* https://www.frontex.europa.eu/media-centre/news/news-release/frontex-and-italy-complete-trial-of-the-largest-drones-the-agency-has-yet-flown-from-a-patrol-ship-wzuanG.

Real-time monitoring: redefining decision-making from sky to ground (and back)

The ability to share real-time video and other data from manned and unmanned aerial assets is the second major innovation introduced by the current aerial bordering regime. As noted above, aircraft flying under MAS live stream video and other data to Frontex's headquarters as well as to relevant national and European institutions, for instance those of countries hosting Frontex Joint Operations.[100]

Critical information about migrants' boats at sea is therefore transmitted via a broad assemblage of communication and visualisation technologies that extend not only across a vast network of airborne sensors, ground stations, and satellite relays, but also across various legislative and jurisdictional frameworks, including maritime law, refugee law, and human rights law.

Far from being a purely technical matter, this data-sharing arrangement has decisive implications for decisions about whether to provide—or, at times, withhold—life-saving assistance to those at sea. It further blurs the already contested allocation of responsibility between the various actors involved, allowing them to evade their obligations by presenting themselves as mere conduits of information who bear no decision-making authority. This dynamic becomes clear when analysing this scattered chain of communication from sky to ground and back again.

For instance, private companies leasing their aircraft to Frontex have consistently stressed that the information gathered by their airborne and ground crews is "directly transmitted to the Frontex command and control centre as well as to centres of the respective Coast Guards" and that they "[do] not take any action in this respect."[101] Yet, various bodies of law, as well as operational manuals, which translate legal obligations into practical procedures, assign ultimate authority for aircraft behaviour to pilots. For instance, recognising a humanitarian obligation to assist persons in distress, the International Aeronautical and Maritime Search and Rescue Manual specifically assigns pilots certain duties, such as that of "inform[ing] the nearest appropriate Rescue Coordination Centre" and of "attempt[ing] to establish communications with the person(s) in distress."[102]

This lack of clarity over the attribution of responsibility persists once the collected data and live video reach the Frontex European Surveillance Room (previously also referred to as the Frontex Surveillance Room, or FSC). While the inner workings and decision-making structures in the room remain largely opaque, various documents offer some insight.

Frontex's Serious Incident Report[103] concerning the shipwreck of February 2023 in Cutro, Southern Italy, reveals, for instance, who sits in the room and how they respond to situations of distress at sea: "in addition to sending

100 Frontex, letter from Leggeri to Oel, "Written Questions Submitted by MEP Demirel," January 14, 2020.

101 Sunderland and Pezzani, Airborne Complicity and Frontex, response to Human Rights Watch, ref. ADNS M22002290, September 14, 2022

102 Relevant sources include the Convention on International Civil Aviation (especially Annex 12), the International Convention on Maritime Search and Rescue, the International Convention for the Safety of Life at Sea (SOLAS) and the International Aeronautical and Maritime Search and Rescue Manual (IAMSAR).

103 The Serious Incident Reporting (SIR) mechanism is the key mechanism for reporting "violations of Fundamental Rights and of EU or international law rules related to the access to international protection and infringements of the Frontex Code of Conduct", as well as other events "which may affect, or be relevant to, the Frontex mission, its image, the safety and security of the participants on the operation." Frontex, "Annex 14: Serious Incident Reporting," n.d., https://www.statewatch.org/media/documents/news/2016/aug/frontex-serious-incident-reporting.pdf.

sighting reports via email", it reads, "Frontex Team Leaders work closely with two Italian experts in the room, who observe the sightings in real-time on the screens and act as liaisons to the International Coordination Centre in Rome and the Italian Maritime Rescue Coordination Centre."[104] Since September 2021, a "Fundamental Rights Officer", who is "mandated with monitoring Frontex's implementation of its fundamental rights obligations in accordance with EU and international law", also sits in the Surveillance Room.[105]

From the response to an EU parliamentary question, we also learn that "considering the international legal framework on search and rescue (SAR), [...] upon detection of a vessel in a situation assessed as life-threatening or having the potential to evolve into a life-threatening situation, [Frontex] notifies of the assessment the competent Rescue Coordination Centre." Frontex maintains that it falls to these Coordination Centres to "state in which SAR phase an incident is (uncertainty, alert or distress according to International Aeronautical and Maritime Search and Rescue Manual)", as well as "all other responsibilities related to SAR incident coordination."[106]

These examples illustrate how responses to "life-threatening" cases are broken down into multiple micro-assessments, each involving different possible operational responses and a multiplicity of actors inside and beyond the Surveillance Room. Decision-making along this scattered chain of communication strategically exploits ambiguity to shift liability onto others, thus enabling the actors involved to evade responsibility for the structural loss of life and human rights violations taking place at sea, despite Frontex's claim that this "enabl[es] an immediate coordinated response."[107]

This pattern has emerged, for instance, in the ongoing public discussions and legal proceedings that have followed recent major shipwrecks, such as the Cutro shipwreck, in which at least 94 passengers lost their lives, and in the Pylos shipwreck south of Crete, where at least 500 people perished. Both incidents occurred after sightings by a Frontex aircraft and in both cases, as noted by the non-governmental group Statewatch, "buck-passing [between the relevant national authorities—Italian in the case of Cutro, Greek in the case of Pylos—and Frontex]. has been prominent." "Frontex stressed", Statewatch continues, "that it provided adequate information to national authorities to prevent the shipwrecks whereas national authorities replied that the information received did not amount to a 'mayday' call to prompt them into immediate action."[108] These incidents highlight the loss of life behind Frontex's fragmented decision-making system and how the latter contributes to a lack of accountability for the deaths and human rights violations taking place at sea.

104 Frontex, "Frontex Incident Reports: Cutro and Pylos," European Border and Coast Guard Agency, accessed September 27, 2026.

105 Jonas Grimheden, Fundamental Rights Officer, Frontex, interview by Chiara Denaro, August 20, 2024; "Fundamental Rights Office," Frontex, https://www.frontex.europa.eu/careers/who-we-are/structure/specific-functions/fundamental-rights-office/.

106 European Commission, Answer to Parliamentary Question E-001757/2022, Annex 2, September 16, 2022, https://www.europarl.europa.eu/doceo/document/E-9-2022-001757-ASW_EN.html.

107 European Parliament, Answer to the written question E-003553/2019 (2019), https://www.europarl.europa.eu/RegData/questions/reponses_qe/2019/003553/P9_RE(2019)003553(ANN)_XL.pdf

108 "From Cutro to Pylos: What Two Shipwrecks Reveal about Europe's Deadly Migration Policies," *Statewatch*, May 22, 2024, https://www.statewatch.org/analyses/2024/from-cutro-to-pylos-what-two-shipwrecks-reveal-about-europe-s-deadly-migration-policies/.

2. Uncovering Patterns: The Role of Aerial Bordering in the "Pushback Machine"

The previous section unravelled the core characteristics of the aerial bordering regime, detailing the means and modalities through which aerial operations are currently conducted. We now turn to a focused analysis of these operations in the creation of what we term the "pushback machine". With this expression we refer to a system in which a pattern repeats: migrants crossing the Central Mediterranean are detected by European authorities, predominantly through their aerial assets; information about their whereabouts is transmitted by the European authorities to their Libyan or Tunisian counterparts, who, in turn, intercept people at sea and forcibly return them to Libya and Tunisia, where they are subjected to widespread abuse.

This "machine" has been widely analysed and denounced by researchers and human rights groups, who have characterised it as a form of "refoulement by proxy" designed to evade human rights responsibility, while maintaining "contactless control" over migrants.[109] Here we seek to unravel the role played specifically by surveillance from the sky within this machine: what it enables, the concrete mechanisms through which it operates and how it has changed interceptions at sea. Our aim here is not so much to adjudicate on the legality of these operational patterns, but to show how they ultimately contribute to systemic abuse.

The Database section of this platform presents all recorded cases of pushbacks contained in Frontex's Joint Operations Reporting Application (JORA) database between 12 August 2018 and 25 January 2023,[110] in which the Agency's planes or drones were directly involved in detecting boats that were subsequently forcibly returned to Libya or Tunisia. This includes 483 cases of pushbacks that took place in the Central Mediterranean, resulting in the forced return of 24,968 people. In 2021 alone, Frontex initiated 36% of interceptions that led to returns to Libya. This should be understood as a lower-bound estimate: recently released data suggest that Frontex was in fact responsible for approximately half of all pushbacks.[111] Additional data produced by the EUNAVFOR Med Irini operation and concerning the period between January 2023 and October 2024 has allowed us to identify an additional 183 cases to the overall count of pushbacks enabled by Frontex aerial surveillance, bringing the total to 611 boats and 33,107 people. While Frontex has refused to share data that would allow us to conduct the same analysis on more recent periods, a recent response to a parliamentary inquiry indicates that the Agency's aerial surveillance program continues to be responsible for half of all pushbacks recorded so far in 2026.[112]

109 Heller and Pezzani, *Mare Clausum*; Violeta Moreno-Lax, "The Architecture of Functional Jurisdiction: Unpacking Contactless Control—On Public Powers, S.S. and Others v. Italy, and the 'Operational Model,'" *German Law Journal* 21, no. 3 (2020): 385–416, https://doi.org/10.1017/glj.2020.25.

110 This does not reflect the timeframe of our overall analysis, but rather the period during which we were able to conduct an in-depth examination of the JORA data obtained by Frontex, a task precluded in earlier and later periods by the Agency's refusal to share certain previously released parts of the dataset. Matthias Monroy, "Seenotrettung: Sea-Watch birgt Ertrunkene im Mittelmeer und kritisiert Frontex," nd, September 29, 2026, https://www.nd-aktuell.de/artikel/1202826.seenotrettung-sea-watch-birgt-ertrunkene-im-mittelmeer-und-kritisiert-frontex.html.

111 Matthias Monroy, "Seenotrettung: Sea-Watch birgt Ertrunkene im Mittelmeer und kritisiert Frontex," nd, September 29, 2026, https://www.nd-aktuell.de/artikel/1202826.seenotrettung-sea-watch-birgt-ertrunkene-im-mittelmeer-und-kritisiert-frontex.html.

112 Matthias Monroy, "Seenotrettung: Sea-Watch birgt Ertrunkene im Mittelmeer und kritisiert Frontex," nd, September 29, 2026, https://www.nd-aktuell.de/artikel/1202826.seenotrettung-sea-watch-birgt-ertrunkene-im-mittelmeer-und-kritisiert-frontex.html.

These figures are not hypothetical but are recorded by the Agency itself. Nevertheless, they likely represent only a fraction of the overall number of forced returns, given the Agency's lack of transparency, the gaps in the datasets we used, and the conservative approach we used in our analysis. A more thorough explanation of the process of obtaining and analysing the JORA data can be read in our Methodology.

The opacity of the agency stands in contrast to the commitments made by Executive Director Hans Leijtens upon taking office, when he pledged to "restore trust by being very transparent about what we are doing and how we are doing it," adding that "there should be absolutely nothing to hide" revealing a widening gap between the agency's public rhetoric113actual disclosure practices.[113]

These numbers alone confirm that aerial bordering is a central pillar of the "pushback machine". Through surveillance, detection, and information sharing, it generates the operational knowledge required to enable pushbacks, while simultaneously shielding European authorities from accountability despite their decisive role in these operations.

The work of numerous researchers, human rights organisations and activist groups confirms the central role that European authorities', and particularly Frontex's, aerial surveillance plays in enabling and facilitating pushbacks. In a statement submitted to the European Ombudsman, Amnesty International found that Frontex's "aircraft have played a key role in identifying the position of boats in the central Mediterranean and its notification to Libyan authorities."[114] Further work conducted by researchers from Liminal in collaboration with Human Rights Watch has also highlighted this point,[115] as have several journalistic investigations.[116]

Building on and deepening this important body of work, the analysis below focuses on three recurring operational patterns that link aerial surveillance to pushbacks. First, Frontex engages in (selective) communication that consistently prioritises facilitating the intervention of Libyan and Tunisian actors over other authorities, even when they are nearby and able to intervene. Second, the intervention of Frontex's aerial assets allows the Libyan Coast Guard to significantly expand its operational reach, leading to interceptions occurring increasingly within the Maltese SAR zone. Third, Frontex aircraft routinely witness the systematically aggressive interception practices of Libyan and Tunisian actors, demonstrating that the Agency knowingly enables and expands their operational role while being fully aware of the violent consequences of their practices. For each pattern, we present a series of exemplary case studies selected from the Database for their relevance and the availability of additional information not included in Frontex's own records. This selection is not meant to be exhaustive: it only includes cases that are paradigmatic and with more documentation available. Incorporating this additional evidence allows for a

113 Bizot, Olivia, Euronews, January 19 2023 "New Frontex chief vows to end illegal pushback s of migrants at border", https://www.euronews.com/2023/01/19/new-frontex-chief-vows-to-end-illegal-pushbacks-of-migrants-at-border

114 Amnesty International, *Contribution to European Ombudsman's Strategic Inquiry OI/3/2023/MHZ: The Role of the European Border and Coast Guard Agency (Frontex) in the Context of Search and Rescue Operations*, https://www.ombudsman.europa.eu/pdf/en/177313.

115 Judith Sunderland and Lorenzo Pezzani, *Airborne Complicity: Frontex Aerial Surveillance Enables Abuse*, Human Rights Watch, December 8, 2022, https://www.hrw.org/video-photos/interactive/2022/12/08/airborne-complicity-frontex-aerial-surveillance-enables-abuse.

116 See: Daniel Howden, Apostolis Fotiadis, and Antony Loewenstein, "Once Migrants on Mediterranean Were Saved by Naval Patrols. Now They Have to Watch as Drones Fly Over," *The Guardian*, August 4, 2019, https://www.theguardian.com/world/2019/aug/04/drones-replace-patrol-ships-mediterranean-fears-more-migrant-deaths-eu; Daniel Howden, Apostolis Fotiadis, and Zach Campbell, "Revealed: The Great European Refugee Scandal," *The Guardian*, March 12, 2020, https://www.theguardian.com/world/2020/mar/12/revealed-the-great-european-refugee-scandal.

better understanding of the human consequences of aerial bordering, and points to the inherent blind spots that emerge when relying exclusively on JORA data.

(Selective) communication

The communication process that follows the detection of a boat at sea is a key mechanism through which aerial bordering contributes to the "pushback machine", as Libyan and Tunisian Coast Guards rely on Frontex communicating the location of migrants' boats at sea. Without this information sharing, interceptions and forced returns by these authorities could not take place. Frontex itself acknowledges that it plays a "vital role" in spotting people in distress at sea and claims that, after a detection, it follows international law and informs the "competent Maritime Rescue Coordination Centre [MRCC]", i.e. the national authority responsible for the rescue activities in a specific area, while also "copying all neighbouring MRCCs".[117] For boats located in the SAR zones of Libya and Tunisia, Frontex treats the Libyan and Tunisian MRCCs as the relevant authorities. However, as human rights organisations have repeatedly argued, the Libyan SAR zone and MRCC are de facto unable to ensure the safety of migrants at sea and exist only as a "legal fiction" created by European authorities to "opportunistically relinquish their duties under the law of the sea" and prevent migrants from reaching Europe by sea.[118]

This dynamic has not always been in place. In a document published in 2017, Frontex explicitly stated that at that time "there [was] no direct operational transmission from Frontex Situation Centre to Libyan Coast Guard forces".[119] A few lines below, the Agency signalled its intention to establish such a channel by soliciting "political guidance from EU institutions" to obtain an authorization "to share sightings with the Maritime Rescue Coordination Centre of Tripoli for the purpose of Search and Rescue."[120]

It remains unclear exactly when Frontex began routinely sharing information with the Libyan Coast Guard. What is certain, however, is that by 2019–2020, human rights organisations were already denouncing the Agency's information-sharing practices as directly facilitating interceptions. While Frontex justified such practices as following maritime law, according to human rights groups this arrangement disregards and violates the prohibition of refoulement under international human rights and refugee law and other fundamental rights obligations.[121] Despite these criticisms, the practice of selective communication has been a consistent feature of aerial bordering.

In 2021, a Freedom of Information request revealed that Frontex officials had sent direct WhatsApp messages and voice notes, including locations and

117 Frontex, letter from Fabrice Leggeri, Executive Director, to Mr. Matthias Oel, "Non-Disclosure by Frontex of the Precise Coordinates of Vessels in Distress at Sea," E-003632/2021, Ares(2021)5106798, August 26, 2021, https://www.europarl.europa.eu/RegData/questions/reponses_qe/2021/003632/P9_RE(2021)003632(ANN01)_XL.pdf.

118 "ASGI Signs the Open Letter to Secretary-General of IMO Requesting the Revocation of Libya's SAR Zone," ASGI, July 6, 2020, https://www.asgi.it/en/news/asgi-signs-the-open-letter-to-secretary-general-of-imo-requesting-the-revocation-of-libyas-sar-zone/; "Mediterranean: As the Fiction of a Libyan Search and Rescue Zone Begins to Crumble," *Statewatch*, May 4, 2020.

119 Frontex, letter from Mr. Fabrice Leggeri, Executive Director, to Mr. Herbert Kickl, "Annual Report on the Practical Application of Regulation (EU) No 656/2014 Establishing Rules for the Surveillance of the External Sea Borders in the Context of Operational Cooperation Coordinated by Frontex," July 16, 2018, 15, https://data.consilium.europa.eu/doc/document/ST-11129-2018-INIT/en/pdf.

120 Frontex, letter from Leggeri to Kickl, "Annual Report on the Practical Application of Regulation (EU) No 656/2014," July 16, 2018, 15.

121 Amnesty International, *Malta: Waves of Impunity*, 2020, https://www.amnesty.org/ar/wp-content/uploads/2021/05/EUR3329672020ENGLISH.pdf.

images, to the Libyan Coast Guard to direct them towards migrants' boats.[122] In early 2023, a visual investigation by *Le Monde* revealed that between 2018 (when Frontex was still officially denying "direct operational transmission") and 2022, at least a dozen images taken by Frontex planes were sent to the Libyan Coast Guard (LYCG).123 A 2024 *Lighthouse Reports* investigation revealed that Frontex had shared the positions of boats with the LYCG more than 2,000 times.[124] These analyses reveal deepening forms of connection and coordination that go well beyond passive information sharing. Moreover, Frontex communication has been on several occasions *selective*: as the Agency frequently fails to inform non-governmental search-and-rescue ships of the presence of a migrant boat in distress, even when these ships are nearby—or even closer to the boat than other actors. In our analysis, we identified 16 such cases in which Frontex aerial assets detected a migrant boat but did not alert nearby non-governmental ships with rescue capacity. When the NGO ships were eventually informed of the boat through other channels, such as the emergency hotline Alarm Phone or non-governmental planes monitoring the Central Mediterranean, they often arrived to find a LYCG vessel in the process of forcibly intercepting and returning the boat's passengers, thus suggesting the LYCG had been informed.

This dynamic should be understood as part of broader trends, outlined in Stopping Civilian Aircraft Monitoring, whereby European authorities have attempted to expel non-governmental ships from the Central Mediterranean through legal and administrative measures, while merchant vessels that are both capable of rescue and bound by maritime obligations to assist boats in distress are often told by MRCCs to "stand by" until the Libyan Coast Guard intervenes, or are even actively instructed to continue their course and disregard distress cases.[125]

It also exposes a broader double standard in the governance of maritime search and rescue. While EU agencies engage in selective communication, humanitarian organisations have faced legal action for failing "to fully communicate information"[126] to all relevant authorities, accusations that have been used to justify vessel impoundments.[127] As it has been noted, this flips the burden of proof: it is now non-governmental groups that must prove that they are abiding by the law, while the actors who are involved in systematic rights violations face little legal scrutiny.[128]

122 European Border and Coast Guard Agency, WhatsApp messages to the Libyan Coast Guard, FOI, FragDenStaat, 2021, https://fragdenstaat.de/anfrage/whatsapp-nachrichten-an-die-libysche-kustenwache/.

123 Arthur Carpentier and Adrien Vande Casteele, "How Frontex Surveillance Equipment Is Used by the Libyan Coast Guard to Intercept Migrants," *Le Monde*, January 9, 2023, https://www.lemonde.fr/en/videos/video/2023/01/07/how-frontex-surveillance-equipment-is-used-by-the-libyan-coast-guard-to-intercept-migrants_6010657_108.html.

124 Bashar Deeb, Steffen Lüdke, and May Bulman, "2,200 Frontex Emails to Libya," *Lighthouse Reports*, February 12, 2024, https://www.lighthousereports.com/investigation/2200-frontex-emails-to-libya/.

125 "Malta Urges Merchant Ships to Ignore Distress at Sea," Sea-Eye, December 19, 2022, https://sea-eye.org/en/malta-urges-merchant-ships-to-ignore-distress-at-sea/.

126 "After Rescuing 65 People: Sea-Eye 5 Unlawfully Detained in Sicily," Sea-Eye, June 17, 2025, https://sea-eye.org/en/after-rescuing-65-people-sea-eye-5-unlawfully-detained-in-sicily/.

127 "Joint Statement: The Obstruction of Search and Rescue Vessels Causes Hundreds of Deaths at Sea," EMERGENCY, July 17, 2025, https://en.emergency.it/press-releases/joint-statement-the-obstruction-of-search-and-rescue-vessels-causes-hundreds-of-deaths-at-sea/.

128 Kiri Olivia Santer, The Borders of Responsibility: Migration Control in the Mediterranean Sea (Duke University Press, 2026), https://www.dukeupress.edu/the-borders-of-responsibility. To try to counteract this imbalance, 13 search and rescue organisations announced the formation of the "Justice Fleet" in November 2025 and simultaneously declared to end operational communication with the Joint Rescue Coordination Centre (JRCC) in Libya. The organisations describe the Libyan Coast Guard as an illegitimate actor at sea and argue that the JRCC cannot be regarded as a competent authority. They point out that the JRCC does not meet international standards, as it is not reachable 24/7, lacks linguistic capacity, and has no adequate technical infrastructure for coordinating rescue operations. See: "Search and Rescue Organisations Suspend Communication with Libyan Rescue Coordination Centre," Sea-Watch, https://sea-watch.org/en/search-and-rescue-organisations-suspend-communication-with-libyan-rescue-coordination-centre/.

Even when the communication is not selective, such as in the case of the Mayday relays issued by Frontex's aircraft on behalf of migrants' boats in distress, it still poses a problem. These emergency messages are broadcast over VHF radio frequencies and can therefore be received by anyone within range of transmission. With the appointment of Hans Leijtens as Executive Director of Frontex in late 2022, the Agency appears to have abandoned its earlier policy of issuing such messages only "in extreme cases"[129] and the number of Mayday relays has indeed increased steadily between 2023 and 2026. However, in a context in which non-governmental actors continue to be systematically expelled from the sea and, when present, are not provided with essential information[130], this increase in Mayday calls appears more as an empty, symbolic gesture, than as a meaningful measure to prioritise rescue over pushbacks.

Frontex tries to deflect criticism by claiming that it communicates the location of boats to Libyan and Tunisian authorities to save lives, and that its cooperation does not go beyond its legal obligation to notify the relevant Rescue Coordination Centre of boats in distress within the Libyan SAR zone. However, this occurs within a context where European authorities have deliberately narrowed the range of possible responses to migrants' boats at sea through specific political and operational choices. The "machine" is deliberately built in a way that makes pushbacks and the forced return of migrants to abuse and detention in Libya and Tunisia the structural outcome of aerial bordering.

This interpretation is corroborated by our Data Analysis, which shows that, according to Frontex's own data, only 2% of operations initiated by a Frontex aircraft spotting a migrant boat between 2016 and 2023 resulted in the passengers being disembarked in Italy or Malta. To characterise the remaining 98% of cases as rescues, as European authorities often do,[131] is not only deceptive but also complicit: by ignoring the systematic abuse to which these 'rescues' lead, such characterisation directly sustains a system that produces violence and death on a large scale.

Extending the Libyan Coast Guard's reach

Another way in which aerial bordering fundamentally contributes to the pushback machine is by extending the Libyan Coast Guard's operational reach. As explored in Chapter 2, Section 1, European authorities have bolstered Libyan and Tunisian forces with political, logistical, and operational support to halt migrant crossings in recent years. Yet, despite the provision of vessels, funding and the training it has received, the Libyan Coast Guard remains largely unable to detect migrants' boats far from the coast. Frontex aerial surveillance fills this gap: while its main objective remains early detection, it can also detect boats and communicate their position even after they have entered the Maltese SAR region, approximately 200km north of the Libyan coast.

The evolution of operations at sea between Libya and Malta reflects the broader trend identified across the Central Mediterranean: the retreat of

129 Fabrice Leggeri to Human Rights Watch, April 25, 2022, quoted in Human Rights Watch, "HRW Submission to the EU Ombudsman's Inquiry into Frontex's Role in the Search and Rescue Operations," statement, December 6, 2023, https://www.hrw.org/news/2023/12/06/hrw-submission-eu-ombudsmans-inquiry-frontexs-role-search-and-rescue-operations.

130 Monroy, Matthias "Frontex withholds details about people in distress – while this data apparently goes to coast guards in Libya", 28 January 2026, https://digit.site36.net/2026/01/28/frontex-withholds-details-about-people-in-distress-while-this-data-apparently-goes-to-coast-guard-in-libya/

131 See: European Commission, letter from Paraskevi Michou, Secretary-General for Migration and Home Affairs, to Fabrice Leggeri, Executive Director, Frontex, March 18, 2019, https://www.statewatch.org/media/documents/news/2019/jun/eu-letter-from-frontex-director-ares-2019%291362751%20Rev.pdf.

European naval assets and the simultaneous expansion of Libyan presence at sea, supported by Frontex surveillance from the sky. For many years, Maltese authorities have been criticised by non-governmental rescue groups for not assisting migrant boats and instead facilitating their pushback in coordination with Libyan militias.[132] This practice intensified after 2020, when Malta and Libya signed a Memorandum of Understanding "in the field of combatting illegal migration". Data from the Malta Migration Archive shows how this led to rescues performed by the Armed Forces of Malta plummeting by 90%, while interceptions by Libyan militias surged by a staggering 230%.[133] Recently leaked documents from the EU naval operation IRINI confirm that Malta continues to "refuse" to participate in migrant rescue missions in its own SAR zone.[134] Meanwhile, Malta has become a central hub for Frontex's aerial surveillance programme, with its Luqa airport acting as the main base for several aircraft as well as the Heron drone and its dedicated ground-control station.

As Maltese non-assistance has continued throughout the past five years, the Libyan Coast Guard patrol boats have increasingly carried out interceptions deeper within the Maltese SAR zone. These vessels have been frequently observed navigating directly from port towards migrant boats, which suggests that they are proceeding to a known location rather than performing randomised patrols. While in some of these cases the position of the migrant boat may have been publicly shared by other sources, in most observed instances, no public alert was issued. This suggests that Frontex has been responsible for communicating the positions of the migrant boats to Libyan authorities in several instances.

An analysis of multiple cases of these LYCG-facilitated pushbacks reveals that these interceptions happen deeper into the Maltese SAR zone. Overall, the Libyan Coast Guard's operational reach has expanded to an area of approximately 99,581 km^2. an expansion likelyfacilitated by Frontex's aerial surveillance and coordination.[135]

132 Amnesty International, *Malta: Waves of Impunity*; "Sabotage, Delays and Non-Assistance: Another Day in Malta's SAR Zone," Alarm Phone, May 3, 2020, https://alarmphone.org/en/2020/05/03/sabotage-delays-and-non-assistance/.

133 Jurgen Balzan, "Malta Migration Archive Exposes Malta's Deadly Refusal to Rescue Migrants at Sea," *Newsbook*, June 10, 2025, https://newsbook.com.mt/en/malta-migration-archive-exposes-maltas-deadly-refusal-to-rescue-migrants-at-sea/.

134 Jacob Borg and James Cummings, "Malta 'Refuses' to Cooperate in Migrant Rescue Missions—Leaked EU Memo," *Times of Malta*, March 13, 2025, https://timesofmalta.com/article/malta-refuses-cooperate-migrant-rescue-missions-leaked-eu-memo.1106464.

135 This figure is not discussed explicitly in the data analysis; rather, it is derived from calculations based on the data points contained in the Malta Migration Archive dataset on interceptions within the Maltese Search and Rescue (SAR) area.

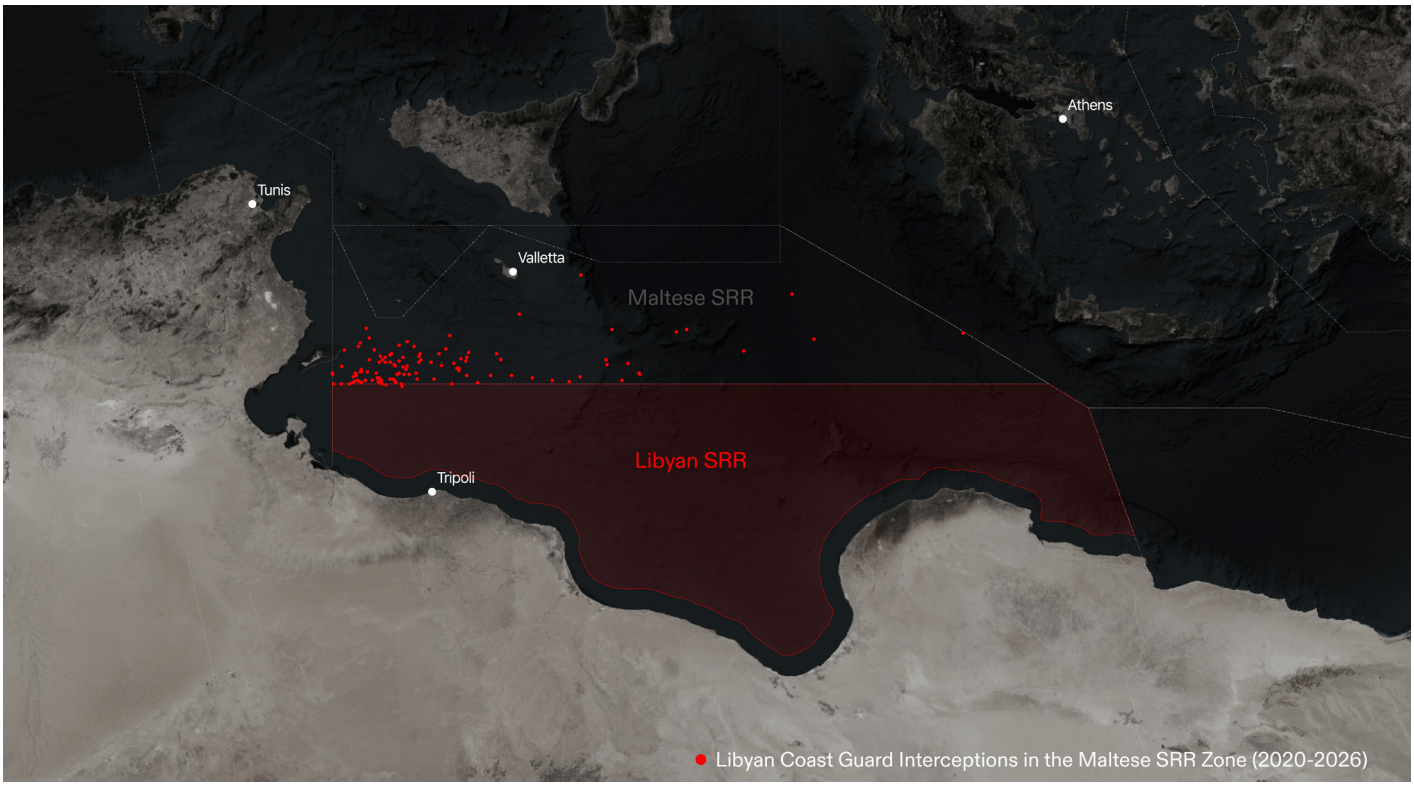


Figure 2.5: Interceptions by the Libyan Coast Guard in the Maltese Rescue Region (2020-2025), data from the Malta Migration Archive

Our analysis has identified at least **13 cases** within the JORA database in which Frontex aerial surveillance detected a boat that was eventually intercepted by the Libyan Coast Guard within the Maltese SAR region, and their passengers forcibly returned to Libya. As noted above, these represent just a few examples of a much wider phenomenon: civil monitoring organisations have frequently documented the Libyan Coast Guard intercepting and returning people to Libya after boats had already entered the Maltese SAR zone. The significance of this specific subset lies in the fact that these cases are recorded in the Agency's own database as involving a Frontex aircraft sighting, as further corroborated by their publicly available flight tracks. This further testifies to the key role played by aerial surveillance in extending the Libyan Coast Guard's reach.

Witnessing direct violence

A third pattern that has emerged reveals that Frontex not only facilitates pushbacks that return migrants to abuse in Libya and Tunisia, but also routinely witnesses, in real time, acts of violence by the Libyan Coast Guard against migrants and non-governmental actors at sea. Search and rescue NGOs have denounced such incidents for years, as both witnesses and frequent victims. Between 2016 and 2026, they have documented at least 76 cases in which the Libyan Coast Guard chased migrants' boats at dangerous high speeds, carried out reckless manoeuvres that endangered lives, physically assaulted and beat migrants, and threatened—often firing—live ammunition at migrants and NGO personnel.[136]

Our analysis identifies a subset of **eight cases** present in Frontex's Joint Operations Reporting Application database in which the Agency's aircraft directly witnessed violent conduct by the Libyan Coast Guard, followed by a pushback. Information from the JORA database has been cross-referenced

136 For a full list of such incidents with more details, see: "Incidents," Justice Fleet, https://justice-fleet.org/en/incidents.

with the corresponding Frontex Serious Incident Reports,[137] which provide important additional details. These reports also contain the assessments by Frontex's Fundamental Rights Officer, who has expressed growing concern and has frequently denounced the non-compliance of Libyan authorities with international and human rights law. Despite mounting evidence of violence and abuse, European actors—particularly Frontex—have continued to expand cooperation with Libyan authorities and to refer to the Libyan Coast Guard's violent interceptions as "rescues".

137 On the Serious Incident Reporting mechanism, see fn. 99 above. See also: Georgios Glouftsios, "Making Pushbacks Public: Secrecy, Material Witnesses and Devices of Dis/Appearance," *Security Dialogue* 55, no. 1 (2024): 3–21; Leonora Karamanidou and Bernd Kasparek, "Fundamental Rights, Accountability and Transparency in European Governance of Migration: The Case of the European Border and Coast Guard Agency Frontex," *Global Migration: Consequences and Responses Working Paper Series*, RESPOND, 2020.

Conclusions

The expansion of aerial bordering described in this report shows no sign of slowing. As aerial surveillance continues to tighten across the Mediterranean, Frontex is simultaneously seeking to expand to other parts of Europe and beyond. In addition to its operations in the Central Mediterranean, the Agency deploys aerial surveillance assets in both the Western and Eastern Mediterranean regions, operating from bases in Greece and Spain. It also conducts extensive monitoring of the EU's eastern borders, the English Channel, and across the Western Balkans, together forming an increasingly dense web of aerial observation. [138]

Frontex is also extending its operational footprint beyond continental Europe. In May 2025, marking a significant milestone for the expansion of its Multipurpose Aerial Surveillance operations, the Agency announced the launch of its first surveillance flights from a non-EU third country, Albania. These missions reportedly "covered both Joint Operations in Albania and Montenegro during a single mission" with "real-time images captured by the aircraft [...] shared with national authorities" of both countries in addition to the Agency's Headquarters in Warsaw.[139] Furthermore, Frontex and Cabo Verde "have launched cooperation on aerial surveillance flights" to monitor maritime routes leading toward the Canary Islands.[140]

To address what it identifies as remaining "gaps" in its surveillance architecture, Frontex is preparing to introduce High Altitude Platform Systems (HAPS) for border monitoring.[141] These long-endurance systems, either aircraft or balloon-based platforms, are capable of remaining airborne for extended periods of time at an altitude of around 20km, occupying the space between the aircraft and the satellites already in use. The Agency's growing interest in persistent, high-altitude monitoring marks a further step towards the occupation of airspaces, aimed at overcoming the "weaknesses" of both satellite and aircraft-based surveillance methods such as limited revisit rates, fuel constraints, and operational downtime, while raising serious legal and human rights concerns.[142]

This suggests that the trends and issues outlined above are unlikely to disappear; if anything, they will intensify and expand. In this sense, aerial bordering offers a unique vantage point to unravel the transformation of bordering practices and to explore the possibilities of challenging border violence more broadly across different contexts. In this conclusion, we briefly point to what we see as some of the most urgent and widely relevant issues emerging from this study.

138 Özlem Demirel, Question for Written Answer, «Frontex Aircraft in Third Countries for the First Time,» E-002099/2025, May 26, 2025, https://www.europarl.europa.eu/doceo/document/E-10-2025-002099_EN.html.

139 Frontex, "New heights for Frontex," Facebook, May 20, 2025, https://www.facebook.com/frontex/posts/998835309094959/.

140 "Frontex and Cabo Verde Launch Aerial Surveillance to Strengthen Maritime Awareness and Save Lives at Sea," Frontex, November 14, 2025, https://www.frontex.europa.eu/media-centre/news/news-release/frontex-and-cabo-verde-launch-aerial-surveillance-to-strengthen-maritime-awareness-and-save-lives-at-sea-c0uEB1.

141 Matthias Monroy, "Frontex Closes Surveillance Gaps in the Air and in Space," *Dissecting Security Architectures*, August 16, 2021, https://digit.site36.net/2021/08/16/frontex-closes-surveillance-gaps-in-the-air-and-in-space/.

142 A research study commissioned by Frontex itself has acknowledged that the deployment of HAPS may raise serious legal and human rights concerns. The study notes that the "risk of abuse of data collected through HAPS [...] might violate the internationally recognised principle of non-refoulement". The report further elaborates that the misuse of surveillance data—for purposes such as facilitating unlawful pushbacks—could constitute a violation of the Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment. "Industry Days on HAPS: High Altitude Platform Systems for Border Surveillance," Frontex, March 13, 2025, https://www.frontex.europa.eu/innovation/announcements/industry-days-on-haps-high-altitude-platform-systems-for-border-surveillance-BOIwkW.

1. The Privatisation of Border Control

Our research into aerial bordering highlights Frontex's role in the *"crafting* of border markets".[143]

It shows that the Agency is not a passive end user of surveillance technologies, simply tapping into existing tools developed by private companies, but that it actively works to generate and develop "markets for border security [...] through tenders, contracts and company relations".[144] Frontex hosts training events and invites manufacturers to present their technological products. Through its Research and Innovation unit, the Agency steers technological development for the surveillance of external borders by "providing and supporting research."[145] These activities have positioned Frontex as a key player in the European homeland security industry. This role is directly enabled by changes in Frontex regulations, which have increasingly etched the transformation of "operational requirements into innovative operational solutions" into the Agency's remit.[146] The new Frontex regulation passed in 2016, in particular, specifies that "the Agency shall *proactively* [...] contribute to research and innovation activities relevant for European integrated border management, including the use of advanced surveillance technology."[147]

This is part of a broader trend in the European and global politics of border control, whereby "the securitization of migration and asylum witnessed in recent decades has been accompanied [and enabled] by rapidly growing commercial interest and activity on *markets for border control*".[148] Through this process, "activities traditionally regarded as central to the state's police power and monopoly on violence [have been] increasingly outsourced to private providers."149 In the context of aerial bordering, this involves not only the procurement of technologies but also the direct involvement of private actors in border control operations. This creates the conditions in which private actors become increasingly entangled in critical decision-making concerning the protection of migrants' rights and lives at sea, such as when pilots monitoring the Central Mediterranean on behalf of Frontex must decide how, what, and to whom to communicate information about a migrant boat.

These companies present themselves as neutral service providers and conduits for information while other actors, namely Frontex or Member States' officials, make the decisions.[150] This narrative, however, needs to be challenged. Technologies, in particular surveillance technologies, are never neutral: they shape what is seen and organise information in ways that tend to direct attention to certain aspects and not others. All technologies are designed with a specific

143 Martin Lemberg-Pedersen, "Crafting Markets of Exclusion: Frontex and the Border Security Sector," *Mondi Migranti*, no. 2 (2024): 9–36, 10, emphasis added, https://doi.org/10.3280/MM2024-002001.

144 "Research and Innovation," Frontex, https://www.frontex.europa.eu/innovation/research-and-innovation/research-and-innovation-at-frontex/.

145 Council of the European Union, Council Regulation (EC) No 2007/2004 of 26 October 2004 establishing a European Agency for the Management of Operational Cooperation at the External Borders of the Member States of the European Union, OJ L 349 (2004): 1, https://eur-lex.europa.eu/eli/reg/2004/2007/oj/eng.

146 Council Regulation (EC) No 2007/2004.

147 European Parliament and Council of the European Union, Regulation (EU) 2016/1624 of the European Parliament and of the Council of 14 September 2016 on the European Border and Coast Guard and amending Regulation (EU) 2016/399 and repealing Regulation (EC) No 863/2007, Council Regulation (EC) No 2007/2004 and Council Decision 2005/267/EC, OJ L 251 (2016): 1, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32016R1624.

148 Lemberg-Pedersen, "Crafting Markets of Exclusion," 13, emphasis added.

149 Mary Bosworth and Lucia Zedner, *Privatising Border Control: Law at the Limits of the Sovereign State* (Oxford University Press, 2022), 1.

150 See Airbus reply to Human Rights Watch, where it claims that "its main mission is 'to support search and rescue missions' and that information it gathers 'is directly transmitted to the Frontex command and control centre as well as to centres of the respective Coast Guards'." Sunderland and Pezzani, *Airborne Complicity*.

purpose in mind and for Frontex—and even more so for the security industries with which the Agency collaborates—that purpose has consistently been enforcing borders and security, not saving lives. As a result, the commercial interests of these companies have been able to shape the scope, modes of operation and scale of EU surveillance.

Moreover, even though various laws place responsibilities on all actors, especially pilots, the growing proliferation of actors, including these private actors, involved in aerial surveillance serves to obfuscate decision-making structures. As a result, it is extremely difficult to hold anyone accountable for the violence and abuse which aerial bordering systematically facilitates.

2. Border Militarisation

Aerial bordering is a key driver of the militarisation of borders.[151] Across the "blurred and overlapping boundaries between Frontex, EU institutions and military-industrial commercial actors," a mutually beneficial relationship has emerged that has made the Agency one of the main agents of border control while expanding the military-industrial complex through public expenditure.[152] Universities, like the one where the authors of this report are based, are also increasingly implicated in border violence through their collaborations with Frontex and the military-industrial complex.[153]

This relationship concerns not only the use of military hardware and tactics but also how bordering practices in the Central Mediterranean are increasingly shaped by military logics, imaginaries and language. This is all the more concerning at a time in which "the boundaries between the economic and the military are becoming ever more blurred" and a "global war regime" has made large increases in public spending for military and security purposes across many countries appear not only legitimate, but even desirable.[154] Similar dynamics can be found in other parts of the world, from the U.S. to Australia and beyond.[155]

This increasing militarisation of borders has been traced back to security concerns emerging from the post-9/11 so-called "War on Terror", the end of the Cold War, as well as colonialism. In tracing these genealogies, several authors have made the important point that it is the very distinction between a supposedly external, military force, and internal police powers that needs

151 Raluca Csernatoni, "Constructing the EU's High-Tech Borders: FRONTEX and Dual-Use Drones for Border Management," *European Security* 27, no. 2 (2018): 175–200.

152 Lemberg-Pedersen, "Crafting Markets of Exclusion," 21–22, emphasis added.

153 Giulia Corgnier, Michele Lancione, Patrícia Nunes Gomes, and Devra Waldman, "Frontex and the University: Positivist Dissonance and the Institutionalisation of Border Violence through Research," *Antipode*, November 2, 2025, https://doi.org/10.1111/anti.70090. It is worth considering also that "in the past two decades, Israeli Aerospace Industries (IAI) has been selected for participation in at least 44 Horizon and other EU research projects, for which it received over €23m—the vast majority of its public money." Privacy International, *Dual-Use Tech: The Israel Aerospace Industries (IAI) Example*, November 10, 2025, https://web.archive.org/web/20251117001454/https://www.privacyinternational.org/report/5703/dual-use-tech-israel-aerospace-industries-iai-example.

154 Michael Hardt and Sandro Mezzadra, "A Global War Regime," *Sidecar*, May 9, 2024, https://newleftreview.org/sidecar/posts/a-global-war-regime.

155 Reece Jones and Corey Johnson, "Border Militarisation and the Re-articulation of Sovereignty," *Transactions of the Institute of British Geographers* 41, no. 2 (2016): 187–200, https://doi.org/10.1111/tran.12115.

to be challenged.[156] Accordingly, "the movement between 'military' and 'civilian' drone applications" that we can also observe in the Central Mediterranean, amongst many other contexts, "cannot be characterized so much as a 'trickle down' from military research and development to nonmartial applications than as part of the infrastructure of an 'ecology of power' that extends beyond the traditional battlefield."[157]

3. Collusion with the (Israeli) Military-Industrial Complex

Within this context, the relationship between aerial bordering and the Israeli military-industrial complex is particularly concerning. As noted above, the drones operated on behalf of Frontex in the Central Mediterranean—the Heron—supplied by Israeli Aerospace Industries (IAI), a 100% state-owned company with close ties to the Technion—Israel Institute of Technology,[158] and the Hermes, supplied by Elbit Systems. Their commercial success is closely tied to being marketed as having been "battle-tested". The Heron drone has been documented as playing a key role in several battlefields, including in successive Israeli assaults on the Gaza Strip since 2008.[159] Since the beginning of Israel's current genocidal campaign, IAI has recorded its "highest-grossing year since the company's inception".[160]

Israeli drones' long-loitering capability, their capacity to remain airborne over a given area for prolonged periods of time, is what makes them particularly suitable for both border surveillance in the Central Mediterranean and enforcing an "aerial occupation" over Gaza.[161] Their persistent presence and the near-constant threat they represent have produced deep psychological damage to the Gazan population, a pattern seen in several other regions that have endured prolonged drone warfare in recent years.[162]

Granting contracts to and using equipment produced by the Israeli defence industry, as Frontex has repeatedly done, offers financial support and global legitimacy to companies directly involved in what a UN Independent Commission of Inquiry, among numerous other leading human rights bodies,

156 Mark Neocleous, *War Power, Police Power* (Edinburgh University Press, 2014); Mark Neocleous, "Air Power as Police Power," *Environment and Planning D: Society and Space* 31, no. 4 (2013): 578–93, https://doi.org/10.1068/d19212. See also: "Everyday Militarisms: Hidden in Plain Sight/Site," ed. Caren Kaplan, Gabi Kirk, and Tess Lea, *Society+Space*, March 8, 2020, https://www.societyandspace.org/forums/everyday-militarisms-hidden-in-plain-sight-site.

157 Caren Kaplan and Andrea Miller, "Drones as 'Atmospheric Policing,'" *Public Culture* 31, no. 3 (2019): 3, https://doi.org/10.1215/08992363-7532679.

158 Privacy International, *Dual-Use Tech: The Israel Aerospace Industries (IAI) Example*; for the origins of IAI in the Technion, see: Maya Wind, *Towers of Ivory and Steel: How Israeli Universities Deny Palestinian Freedom* (Verso, 2024).

159 "Heron," The Database of Israeli Military and Security Export, https://dimse.info/heron/; Privacy International, *Dual-Use Tech: The Israel Aerospace Industries (IAI) Example*.

160 "IAI Publishes Its Annual Financial Statements 2023," Israel Aerospace Industries, March 14, 2024, https://www.iai.co.il/press/iai-publishes-its-annual-financial-statements-for-2023/.

161 Hellinger and Bhavnani, "Aerial Occupation and Aerial Forensics in Gaza"; Emery and Brunstetter, "Drones as Aerial Occupation."

162 Nasser Hussain, "The Sound of Terror," *Boston Review*, October 16, 2013, https://www.bostonreview.net/articles/hussain-drone-phenomenology/; International Human Rights and Conflict Resolution Clinic at Stanford Law School and Global Justice Clinic at NYU School of Law, *Living Under Drones: Death, Injury, and Trauma to Civilians from US Drone Practices in Pakistan*, 2012, https://law.stanford.edu/international-human-rights-and-conflict-resolution-clinic/projects/living-under-drones/; Mattia Barro, "La 'Guerra sonora' dei droni," *Rolling Stone Italia*, December 10, 2024, https://www.rollingstone.it/cultura/la-guerra-sonora-dei-droni/954659/.

has found to amount to a genocide.[163] This support strengthens Israel's capacity to continue committing war crimes in Gaza and implicates the EU in these violations, alongside the direct responsibility of the European border regime, which has contributed to the death of many attempting to flee from Gaza since the onset of the brutal Israeli blockade in 2007.[164] Heron drones belonging to the Greek Coast Guard have also been documented to have participated in the surveillance of both the Freedom Flotilla[165] in June and the Global Sumud Flotilla in September 2024 while they were transiting south of Crete on their way to Gaza.[166]

Israeli companies and their partners are fully aware of the value of these Frontex contracts and have repeatedly publicly acknowledged them. As early as the first test flights for Frontex in 2018, an IAI representative stated in a press release that "flying in European airspace is a breakthrough for IAI. [...] We believe this is a significant step forward for IAI into other commercial markets for the Heron UAS [unmanned aerial system]." This sentiment was echoed by the CEO of the Airbus subsidiary involved in this contract, who emphasised: "This long endurance maritime surveillance contract marks a new milestone in our ability to extend our successful unmanned operational services from military customers to civilian stakeholders such as FRONTEX and eventually other EU agencies."[167] Upon signing in 2020 its first contract that followed earlier initial trials, an IAI Executive Vice President stressed again that the contract with Frontex "will open the door to more civilian markets."[168]

While the export of offensive weapons to Israel has come under increased legal scrutiny and has been the target of various forms of direct action, the inward flow of arms from Israel to Europe has attracted far less attention.[169] Earlier campaigns to stop the use of military technologies implicated in Israeli war crimes by EU agencies should be revisited in light of the ongoing genocide in Gaza and the enduring apartheid regime enforced by Israel.[170]

163 "Israel Has Committed Genocide in the Gaza Strip, UN Commission Finds," OHCHR, September 16, 2025, https://www.ohchr.org/en/press-releases/2025/09/israel-has-committed-genocide-gaza-strip-un-commission-finds.

164 See: Peter Beaumont and Patrick Kingsley, "Devil and the Deep Blue Sea: How Mediterranean Migrant Disaster Unfolded," *The Guardian*, October 1, 2014, https://www.theguardian.com/world/2014/oct/01/-sp-sea-mediterranean-migrant-disaster.

165 Gaza Freedom Flotilla, "The use of Heron drones by Greece...", Instagram, June 4, 2025, https://www.instagram.com/p/DKfbXoTSlUG/.

166 Riccardo Antoniucci, "C'era un drone della guardia costiera greca sopra le barche della Sumud Flotilla, la notte dell'attacco. Ecco il tracciato", *Il Fatto Quotidiano*, September 24, 2025, https://www.ilfattoquotidiano.it/2025/09/24/sumud-flotilla-drone-guardia-costiera-greca-attacco-notizie/8138054/.

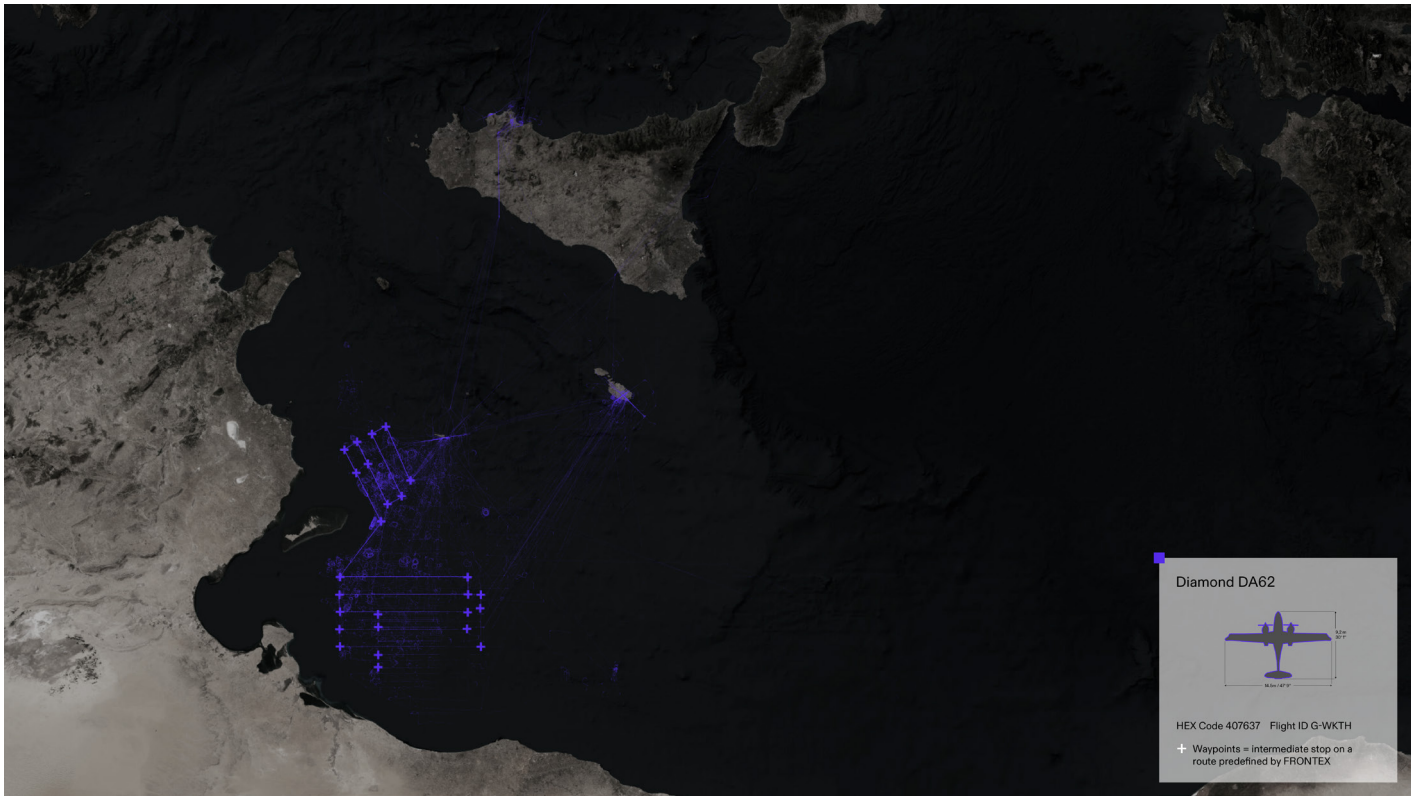
167 "IAI and Airbus Maritime Heron Unmanned Aerial System (UAS) Successfully Completed 200 Flight Hours," Israel Aerospace Industries, https://www.iai.co.il/iai-and-airbus-maritime-heron-unmanned-aerial-system-uas-successfully-completed-200-flight-hours.

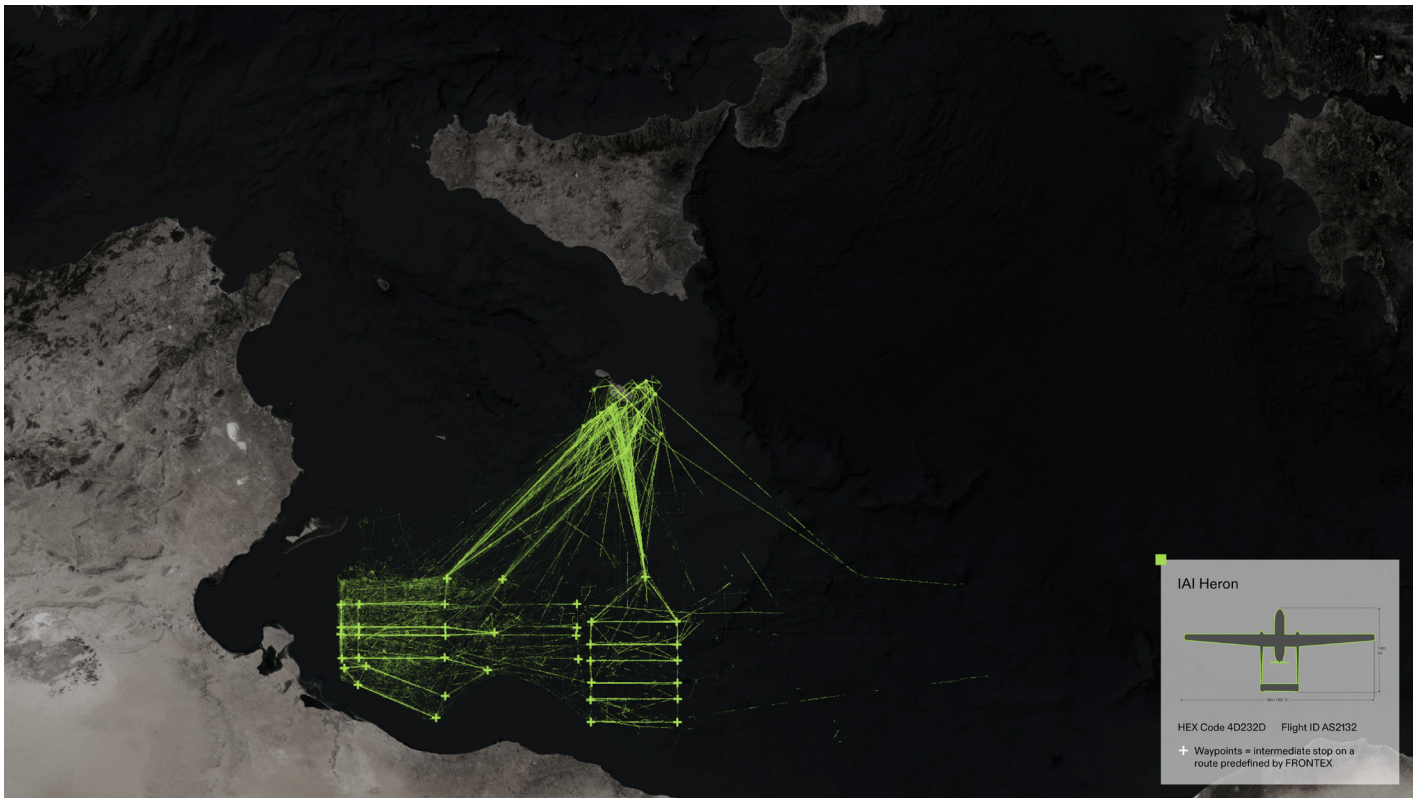
168 European Border and Coast Guard Agency (Frontex) selects Airbus and its partner IAI for Maritime Aerial Surveillance with Remotely Piloted Aircraft Systems (RPAS)," Airbus, October 20, 2020, https://web.archive.org/web/20250211161724/https://www.airbus.com/en/newsroom/press-releases/2020-10-european-border-and-coast-guard-agency-frontex-selects-airbus-and

169 Alon Sahar and Itamar Mann, "The Other Side of Trade," *Verfassungsblog*, August 26, 2025, https://verfassungsblog.de/the-other-side-of-trade/.

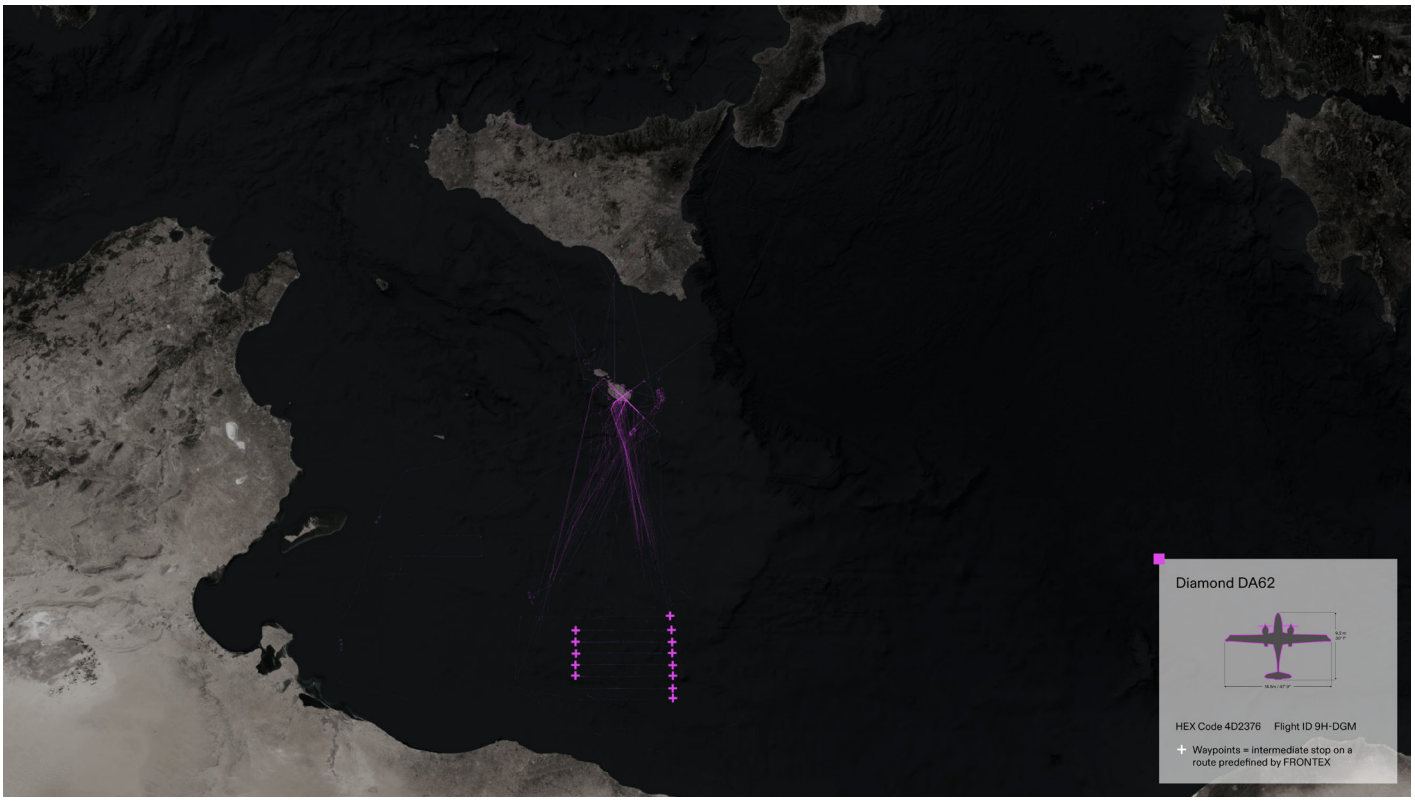
170 "Stop the Wall Petition," Stop the Wall, https://petition.stopthewall.org/.

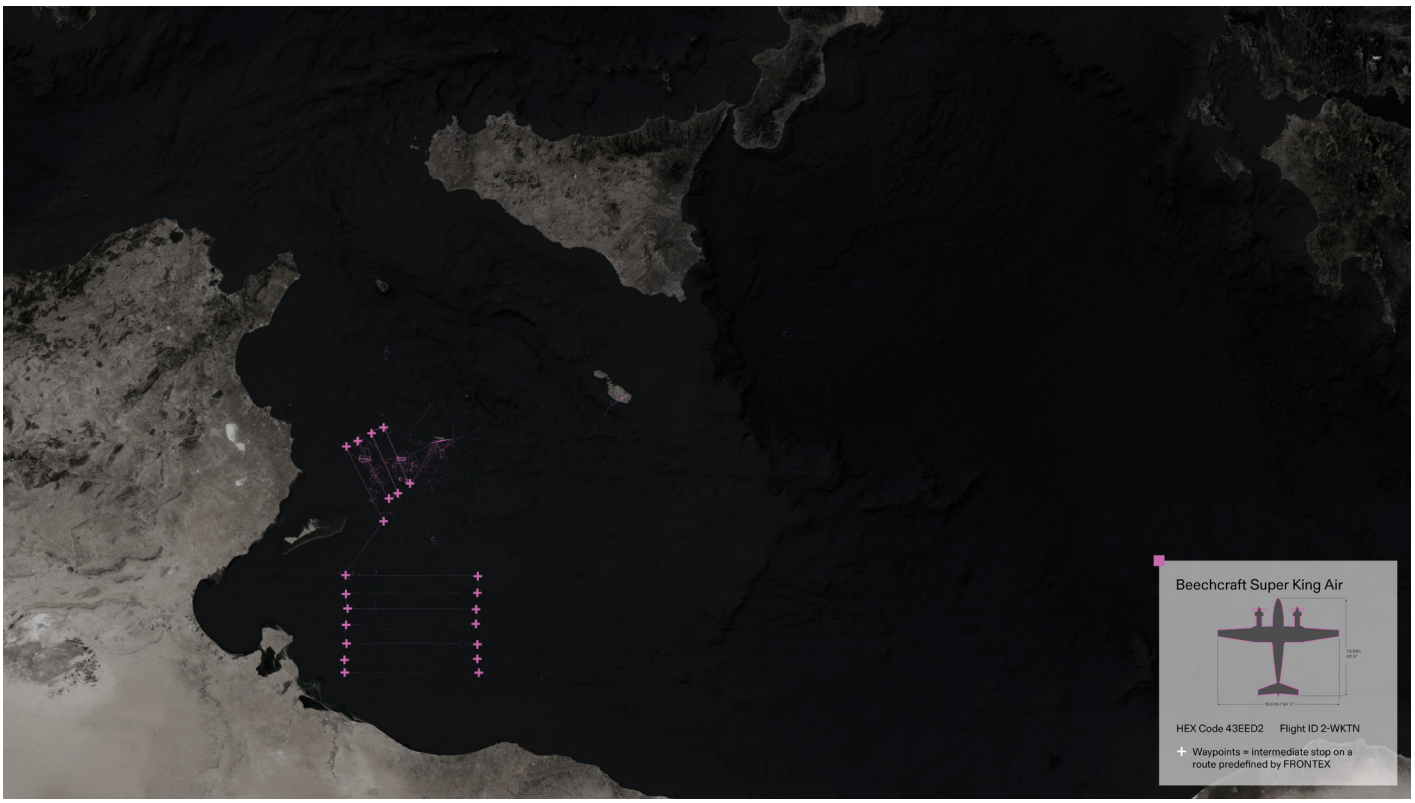
Annex





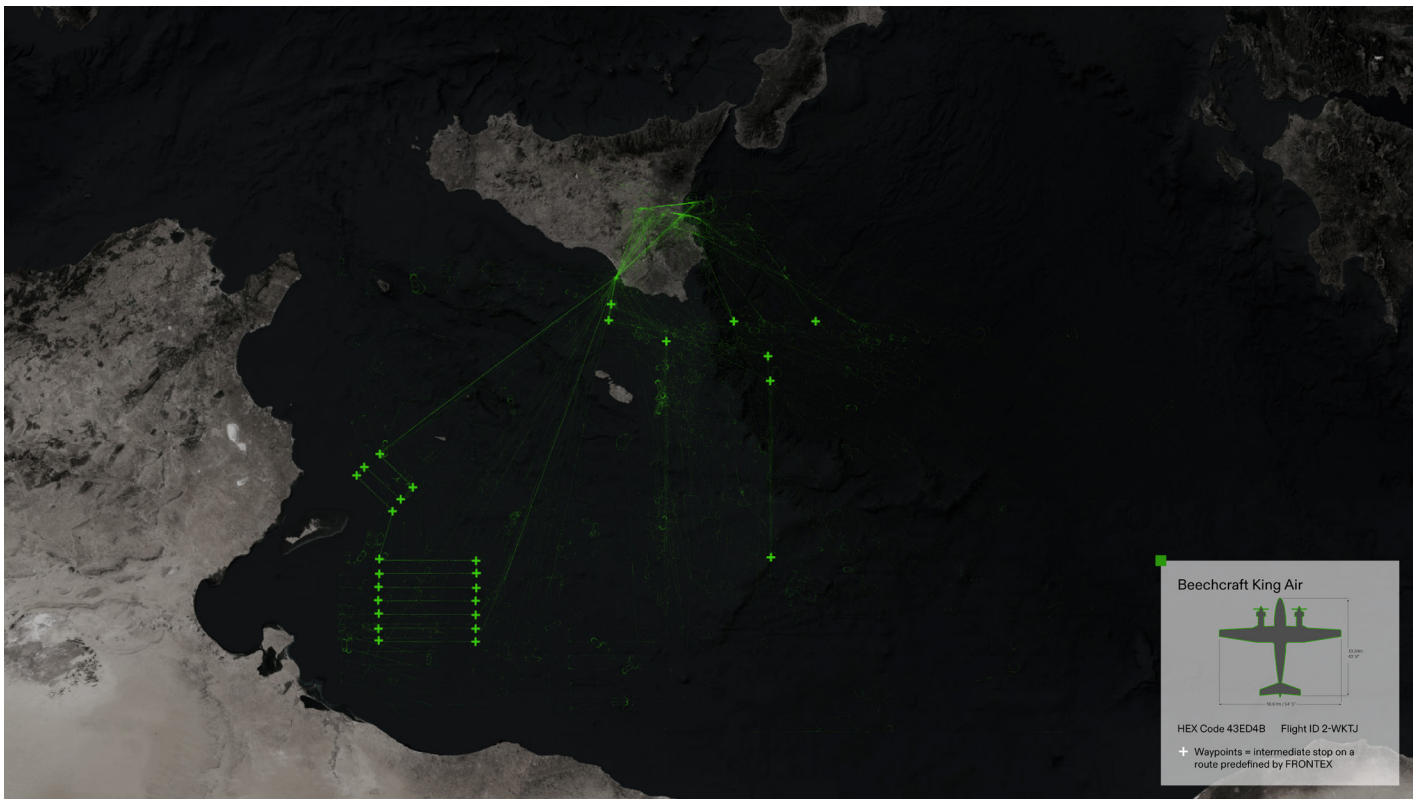
Annex images 1-2: all waypoints by Frontex aerial assets seperated by asset





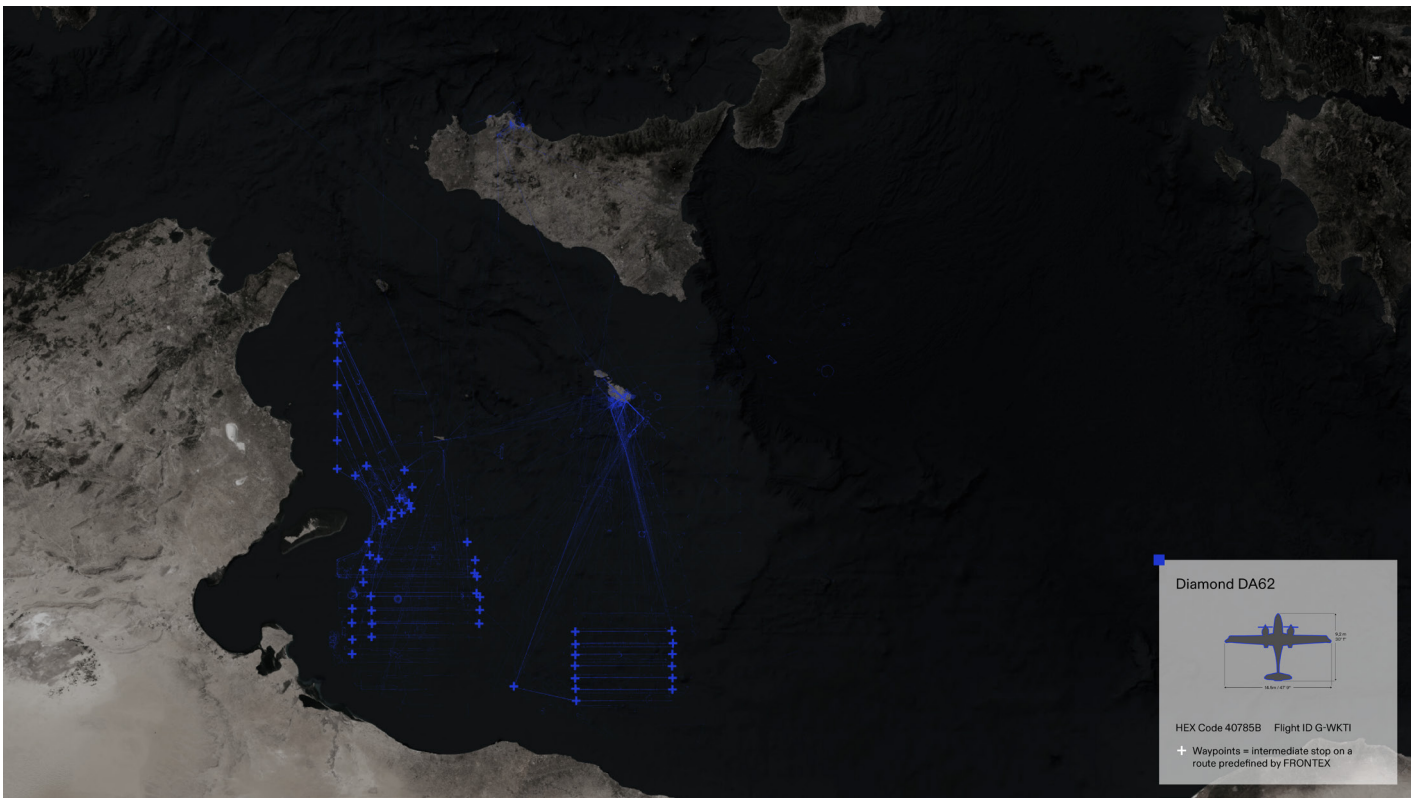
Annex images 3-4: all waypoints by Frontex aerial assets seperated by asset





Annex images 5-6: all waypoints by Frontex aerial assets seperated by asset





Annex images 7-8: all waypoints by Frontex aerial assets seperated by asset